
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

Anonymous Authors

ABSTRACT

Uniform manifold approximation and projection (UMAP) has become one of the most widely adopted neighbor embedding methods, generating its low-dimensional representations through the interplay of attractive and repulsive forces acting on high-dimensional data points. In this work, we decompose these forces into their constituent *attraction* and *repulsion shapes* and, on the basis of this decomposition, establish precisely how each governs cluster formation and visualization across the broad family of attraction–repulsion embedding methods. Repulsion accentuates dissimilarity and thereby controls cluster boundaries and inter-cluster separation, whereas attraction operates far more subtly: the attractive tension between a pair of points manifests simultaneously as contraction and as expansion in the low-dimensional mapping. This dual character accounts for the necessity of learning-rate annealing and explains why the attractive and repulsive terms warrant distinct treatment. Exploiting this understanding, we modify the attraction shape and improve the consistency of cluster formation under random initialization, providing direct evidence that UMAP encodes a unique underlying structure. Taken together, our analysis renders UMAP and related embedding methods appreciably more interpretable, more robust, and more accurate.

1 Introduction

The contemporary scientific landscape is defined by a pervasive abundance of high-dimensional data, and dimensionality reduction (DR) has consequently become an indispensable tool for exploratory analysis, rendering the latent structure of such data visually accessible. The two most widely adopted methods, *t*-distributed stochastic neighbor embedding (Maaten and Hinton, 2008) and uniform manifold approximation and projection (UMAP) (McInnes et al., 2018), are both grounded in the same attraction–repulsion dynamics, drawing similar data points together while driving dissimilar ones apart. Being unsupervised, these algorithms dispense with labeled data entirely; they instead recover and preserve the intrinsic structure of high-dimensional data by exploiting local (attractive) and global (repulsive) relations among points. This label-free formulation makes them exceptionally well suited to a broad range of tasks in which visualizing latent patterns in unlabeled data is both critical and informative, including clustering (Becht et al., 2019), exploratory data analysis (Fleischer and Islam, 2020), anomaly detection in semiconductor manufacturing (Fan et al., 2021), visual search (González-Márquez et al., 2024), time series analysis (Altin and Cakir, 2024), the study of representation convergence (Huh et al., 2024), and outlier image detection (Islam and Fleischer, 2024). By learning embeddings in a data-driven, label-free manner, DR epitomizes the capacity of unsupervised methods to distill complex data into interpretable form.

Building on this same attraction–repulsion principle, a succession of newer methods has emerged (Amid and Warmuth, 2019; Agrawal et al., 2021; Wang et al., 2021; Narayan et al., 2021; Yang et al., 2022; Wang et al., 2025; Kury et al., 2025), each engineered to accentuate particular aspects of the data. Yet, notwithstanding their demonstrated utility across diverse applications, these methods rest heavily on heuristic practices that frequently resist meaningful interpretation, and DR inevitably introduces distortions that cannot be eliminated (Chari and Pachter, 2023). A principled, mechanistic understanding of these algorithms is therefore not merely desirable but essential: it is what enables practitioners to interpret embeddings correctly, to discount spurious structure, and to optimize performance deliberately rather than by trial and error. In practice, compact clusters are obtained through an assortment of devices—bespoke initialization, learning-rate schedules, and kernel tuning—yet the underlying dynamics of the attractive and repulsive forces that actually drive cluster formation have never been examined in depth. Compounding the difficulty, the parameters that

genuinely matter remain buried within abstract functional forms, which obstructs any transparent account of algorithmic behavior.

In this paper, we resolve this opacity by decomposing the embedding forces into their constituent parts and extracting the functional *shapes* of attraction and repulsion. This decomposition yields a unifying result: the necessity of learning-rate annealing, the difficulty of producing consistent output under random initialization, and the very origin of cluster formation all trace back to attraction, whereas repulsive forces principally govern inter-cluster distances. Recasting the forces as functional shapes thus turns a collection of heuristic observations into a coherent and predictive account of embedding dynamics. Our specific contributions are as follows:

1. We introduce the attraction and repulsion *shapes*—the distance-dependent scalar coefficients obtained by decomposing the attractive and repulsive gradients—and establish exact per-iteration conditions under which each induces contraction or expansion of distance, furnishing a general and analytically tractable perspective on this entire class of algorithms (Section 4).
2. We demonstrate that UMAP’s attraction shape produces the counterintuitive phenomenon whereby a single attractive update yields both contraction and expansion of distance, and, by comparing attraction shapes across algorithms, we identify this mechanism as the direct cause of UMAP’s dependence on learning-rate annealing (Section 5).
3. We modify the attraction shape to markedly improve the consistency of embeddings under random initialization, thereby providing concrete evidence that UMAP encodes a unique underlying structure (Section 5.1).
4. Through analysis of the repulsion shapes, we establish that attraction governs cluster formation while repulsion regulates cluster compactness and inter-cluster distance (Section 5.2).
5. We compare the attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG-*t*-SNE, expose their precise similarities and distinctions, prove that NEG-*t*-SNE and Parametric UMAP are equivalent, and characterize the stability of each algorithm (Section 6).

We center the main text on UMAP—now the de facto standard across many fields—and introduce the other algorithms where they sharpen the analysis; the appendix treats each in full detail. The code used in this research will be released upon acceptance.

2 Related Works

Modern iterative graph-based neighbor embedding algorithms originate with stochastic neighbor embedding (SNE) (Hinton and Roweis 2002) and its heavy-tailed extension, *t*-SNE (Maaten and Hinton 2008), which replaces the Gaussian low-dimensional kernel with a *t*-distribution. Both construct a dense graph in which every point maintains a pairwise relation to every other, irrespective of mutual similarity, and both normalize the graph weights so as to induce a probability distribution. Concurrent methods—locally linear embedding (Roweis and Saul 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi 2002)—instead operate on a *k*-nearest neighbor (*k*-NN) graph of pairwise interactions; as spectral methods, they obtain the embedding through eigenvalue decomposition. Tang et al. (2016) subsequently embedded the *k*-NN graph within the iterative optimization framework and dispensed with normalization in the low dimension. McInnes et al. (2018) carried this reduction to its conclusion in UMAP, eliminating normalization entirely—in both high and low dimensions—and computing the embedding from pairwise interactions alone. The resulting optimization applies an explicit attractive force to preserve local neighborhoods and an explicit repulsive force to hold dissimilar points apart. Building on these foundations, methods such as PaCMAP (Wang et al. 2021) and NEG-*t*-SNE (Damrich et al. 2023) have since been proposed; for a recent survey, see de Bodt et al. (2025).

A substantial body of work has sought to understand and explain the relationships among these algorithms. An early analysis of SNE established that data well clustered in the original space remain well clustered in the embedding space (Shaham and Steinerberger 2017). The corresponding analysis of *t*-SNE by Linderman and Steinerberger (2019) showed that the number of clusters in the embedding space is a lower bound on the number of clusters in the original space, a result subsequently refined by further theoretical characterizations (Arora et al. 2018; Cai and Ma 2022; Linderman and Steinerberger 2022). Because *t*-SNE and UMAP descend from a common framework yet yield strikingly different visualizations, a central line of inquiry has been to pin down the connection between them (Böhm et al. 2022; Damrich et al. 2023; Draganov et al. 2023). Böhm et al. (2022) posited that Laplacian Eigenmaps, ForceAtlas2 (Jacomy et al. 2014), UMAP, and *t*-SNE are all samples drawn from a single underlying spectrum; LE and *t*-SNE are linked through the early-exaggeration phase (Cai and Ma 2022), and UMAP and *t*-SNE are related through contrastive estimation (Damrich et al. 2023). Contemporaneously, Hu et al. (2023) independently uncovered the connection between contrastive learning and SNE. More recent accounts explain dimensionality reduction through a probabilistic

perspective (Ravuri et al. 2023, Ravuri and Lawrence 2024), kernel techniques (Draganov and Dohn 2023), and information geometry (Kolpakov and Rocke 2024). Our shape-based decomposition is complementary to these efforts: rather than relating whole algorithms, it isolates the two functional primitives—attraction and repulsion—through which their differences can be read off directly.

3 Uniform Manifold Approximation and Projection

UMAP represents the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ as a high-dimensional graph whose edges encode a pairwise relation $p_{i,j} = f_h(d(x_i, x_j))$, normalized to lie in $[0, 1]$ (see Appendix H for the full construction).

The low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is endowed with an analogous graph defined by the differentiable affinity function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^b)^b}, \quad (1)$$

in which the parameters a and b control the density of the mapping and are determined by fitting q_{ij} to the piecewise target

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d, \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

whose parameter m_d fixes the minimum admissible distance between the two nearest low-dimensional points.

UMAP then obtains the embedding by minimizing the cross-entropy loss

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

Here the first term generates an attractive force and the second a repulsive force. Rather than updating every pair at each iteration, UMAP adopts the negative-sampling scheme (Mikolov et al. 2013, Tang et al. 2016): for each edge with $p_{ij} > 0$ —a *positive edge*—a small set of *negative edges* is drawn at random. The attractive force acts on the positive edge,

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

whereas the repulsive force acts on the negative edges,

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t indexes the optimization step. Crucially, y_j is left unchanged for negative edges—an asymmetry that will prove central to the analysis that follows. For a detailed treatment of UMAP’s loss, see (Damrich and Hamprecht 2021).

4 Attraction and Repulsion Shapes

The updates in Eqs. (4)–(6) admit a decisive simplification: each gradient $\nabla_{y_i^t} \log(q_{ij})$ (respectively $\nabla_{y_i^t} \log(1 - q_{ij})$) factors into a scalar coefficient that depends solely on the distance $\|y_i - y_j\|_2$, multiplying the displacement vector $(y_i - y_j)$. We designate this scalar coefficient the attraction (respectively repulsion) *shape*. Although we develop the formalism using UMAP as the running example, it applies without modification to any method built on attraction and repulsion.

Writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$ with $\zeta = \|y_i - y_j\|_2$, the update of a positive edge takes the form

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

where $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape and we have used the identity $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$, valid under the Euclidean metric. The update of a negative edge is reformulated analogously as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Decompositions of this general kind have appeared before (McInnes et al., 2018; Agrawal et al., 2021; Draganov and Dohn, 2023), but their formulation and use differ fundamentally from ours. The original UMAP paper (McInnes et al., 2018) invoked it as a computational trick to accelerate processing, while (Draganov and Dohn, 2023) employed it to compare algorithms from a kernel standpoint; in both cases the goal was simply to evaluate the derivative. We instead treat the two factors as independent functions in their own right—objects that can assume a variety of forms and be analyzed, compared, and deliberately modified. (Agrawal et al., 2021) took a related step earlier, but expressed the decomposition as a scalar coefficient times a *unit* vector $(y_i - y_j)/\|y_i - y_j\|_2$, thereby foregrounding force magnitude alone. As the analysis below makes clear, our shape decomposition is considerably more illuminating than magnitude, because it retains precisely the distance dependence that dictates whether a pair contracts or expands.

4.1 Conditions for Attraction and Repulsion

We now derive, directly from the update equations (7)–(10), exact conditions under which the shapes contract or expand pairwise distances. The following proposition characterizes contraction of the distance between the pair y_i^t and y_j^t of a positive edge.

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

The product λf_a thus acts as the *effective* attraction shape; in most cases, however, f_a alone suffices to draw meaningful conclusions about the resulting embeddings. The companion proposition characterizes expansion of the distance between the pair y_i^t and y_j^t of a negative edge.

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

Including a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ in Eq. (10) leaves this conclusion intact. We emphasize that these conditions constitute a per-iteration *certificate* of guaranteed contraction or expansion, and must not be conflated with learning-rate tuning or decay: the shapes can be designed so that contraction and expansion hold independently of any learning-rate schedule.

Figure 1(a) illustrates the effect of representative values of $f_a < 0$ and $f_r > 0$ on a pair of points. The repulsion shape behaves straightforwardly: any positive value enlarges the distance. The attraction shape is far more subtle, because f_a encodes both attractive and repulsive dynamics within a single sign convention. For $f_a \in (-1.0, 0)$ the distance decreases, attaining maximal attraction at $f_a = -0.5$, where the pair becomes coincident and the displacement sign flips; any value below -1.0 instead enlarges the distance. Although each shape acts on an individual pair, their aggregate effect is what sculpts the global structure of the embedding.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Applying the gradient decomposition to the low-dimensional affinity (1) yields UMAP’s attraction and repulsion shapes in closed form,

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section examines UMAP’s default shapes, which are governed principally by the parameters a and b , and extracts the insights obtained by perturbing them. The analysis holds for $b \leq 1$, the regime in which the attraction shape is strictly increasing and hence invertible (see Appendix A for the derivation).

Figure 1(b) displays UMAP’s default attraction shape ($a = 1.58, b = 0.89$), which diverges to ∞ as $\zeta \rightarrow 0$. As Proposition 4.1 predicts, the transition from contraction to expansion occurs precisely where λf_a^U crosses -1 while ζ approaches 0. Because the shape is invertible, the transition distance $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$ is well defined and constitutes the *minimum distance for contraction* under attractive updates: pairs with $\zeta > \zeta_{-1}$ contract, whereas pairs with $0 < \zeta < \zeta_{-1}$ expand. This directly contradicts the prevailing intuition that attractive updates invariably draw points together, and it is the mechanistic crux of the analysis that follows.

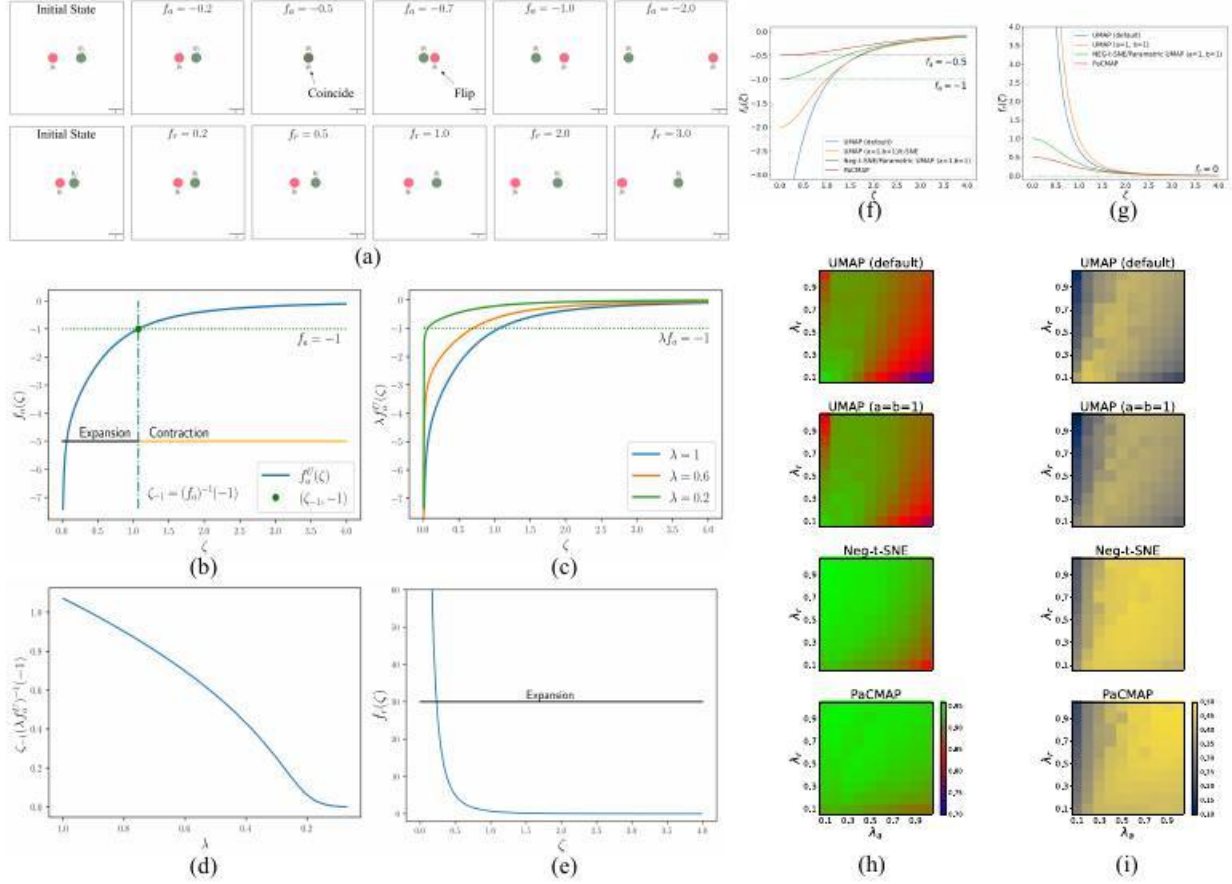


Figure 1: Attraction and repulsion shapes in UMAP. (a) Effect of different values of the attraction shape f_a (top) and repulsion shape f_r (bottom) on a pair of points. (b) Attraction shape of UMAP, which diverges as $\zeta \rightarrow 0$. (c) Effective attraction shape λf_a across learning rates λ , showing that decreasing λ confines the shape within the contraction regime. (d) Minimum distance for contraction ζ_{-1} decreasing toward zero as λ decreases. (e) Repulsion shape of UMAP. (f) Attraction and (g) repulsion shapes of several embedding methods, isolating where UMAP differs from its successors. (h) Trustworthiness and (i) Silhouette Score of the methods as the constant learning rate is varied independently for attraction (λ_a) and repulsion (λ_r). Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

When ζ_{-1} is large, neighboring points oscillate between contraction and expansion and the clusters appear diffuse. The objective of optimization can therefore be recast in a single, sharp criterion: driving $\zeta_{-1} \rightarrow 0$ so as to obtain crisp cluster boundaries. This requirement explains, rather than merely describes, why UMAP’s learning-rate schedule must anneal to zero (Figs. 1(c,d)). The repulsion shape (Eq. 12), by contrast, is strictly positive and therefore satisfies Proposition 4.2 unconditionally, decaying to 0 as $\zeta \rightarrow \infty$ and diverging as $\zeta \rightarrow 0$ (Fig. 1(e)).

Comparing attraction shapes across algorithms reveals a clear taxonomy: only UMAP and t -SNE must contend with $f_a < -1$ (equivalently $\zeta_{-1} > 0$ at $\lambda = 1$; Fig. 1(f)). t -SNE mitigates this by weighting each update with the corresponding p_{ij} , whereas UMAP relies exclusively on learning-rate annealing. NEG- t -SNE and PaCMAP, by construction, keep f_a within $[-1, 0]$ and thus satisfy Proposition 4.1 for every $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$; PaCMAP’s weighted attraction (< 0.5 throughout) goes further still, precluding sign flips during the attractive update. The repulsion shapes exhibit a parallel dichotomy: only UMAP admits arbitrarily large values at small distances (Fig. 1(g)); the shape is formally unbounded, though its values are typically clipped during optimization). Appendix G details these and related methods.

We quantify the consequences of these attraction/repulsion choices on the MNIST embedding by assigning separate constant learning rates to attraction (λ_a) and repulsion (λ_r) and sweeping each independently (Figs. 1(h,i)). For UMAP, small λ_a (below 0.5, hence $\zeta_{-1} \simeq 0$) is decisively preferable, improving both structure preservation—measured by trustworthiness (T) (Venna and Kaski [2001], Fig. 1(h))—and cluster separation—measured by the silhouette score (SIL) (Rousseeuw [1987], Fig. 1(i))—whereas the full range of $\lambda_r \in [0, 1]$ is admissible. For NEG- t -SNE and PaCMAP,

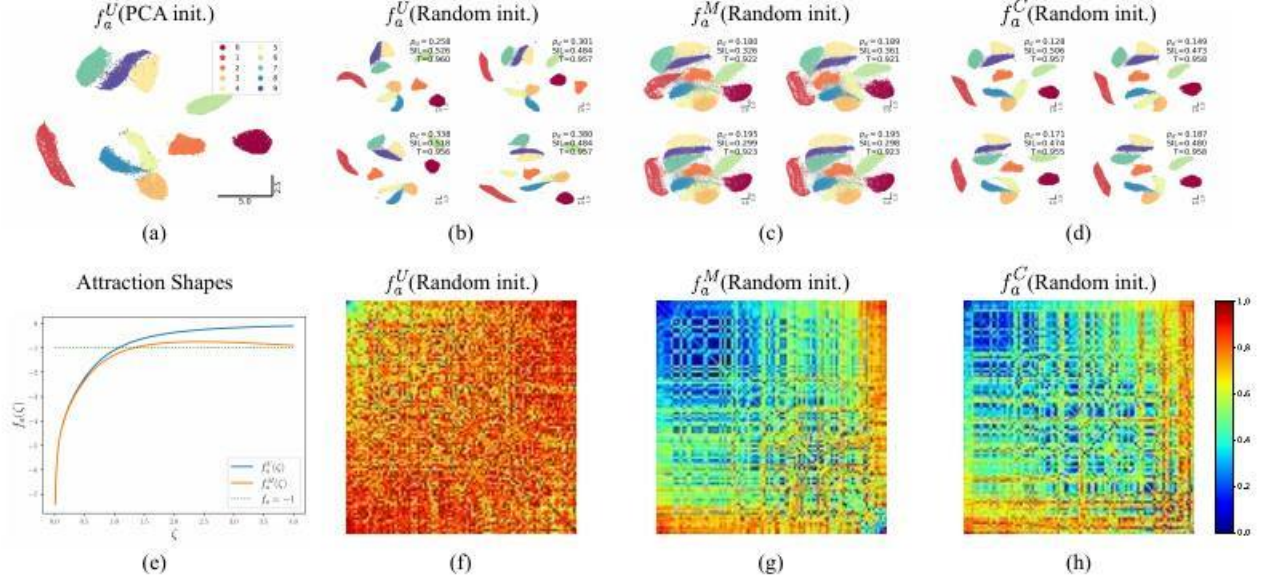


Figure 2: Effect of random initialization on different attraction shapes for the MNIST dataset. (a) Reference mapping obtained with PCA initialization. (b–d) The four mappings with the lowest Procrustes distance (p_d) from the reference in (a), for the (b) UMAP, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f–h) Procrustes matrices aggregated over 100 runs of the (f) UMAP (0.78 ± 0.13), (g) modified (0.49 ± 0.21), and (h) composite (0.50 ± 0.20) attraction shapes. Each diagonal entry (i, i) is sorted by p_d from the reference (a), while each off-diagonal entry (i, j) reports the p_d between the i^{th} and j^{th} mappings. Both the matrices and the reported mean $p_d \pm \text{std}$ establish that UMAP’s embeddings are not self-similar across runs, whereas the modified and composite attraction shapes reduce the mean p_d substantially and enforce a scale-invariant structure.

which already satisfy $\zeta_{-1} = 0$ by construction, the entire parameter range is effective. These sweeps confirm the central claim of this section: UMAP attains satisfactory embedding and clustering only by forcing ζ_{-1} toward zero—a condition it achieves in practice through learning-rate annealing—whereas the newer algorithms make $\zeta_{-1} = 0$ their default. Section 6 and Appendices B, G, and I provide further discussion, results, and illustrations across additional datasets.

5.1 Improving UMAP’s Consistency under Random Initialization

The consistency of UMAP embeddings is well known to hinge on proper initialization (Kobak and Linderman, 2021; Wang et al., 2021). Conventionally, the embedding is seeded by principal component analysis (PCA) of the data or by spectral decomposition of the high-dimensional graph, which yields reproducible mappings despite the various sources of stochasticity in the optimization. Under random initialization, by contrast, clusters frequently fail to form altogether or emerge in an arbitrary orientation on each run. Our shape decomposition identifies the precise cause: when the initial separation between two high-dimensional nearest neighbors is large, the attractive force is too weak to draw them together. This phenomenon, termed *near-sightedness* (Wang et al., 2021), is manifest in the attraction shape, whose magnitude $|f_a^U|$ vanishes with increasing distance ($\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)|\zeta = 0$).

This diagnosis immediately suggests a remedy: inducing “far-sightedness” by amplifying attraction at large distances, so that distant neighbors are still drawn together. To test this prediction directly, we augment UMAP’s attraction shape to strengthen the attractive force,

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where β regulates the strength of the added term (we set $\beta = 0.2$; Fig. 2(e)). This modification of the attraction shape corresponds exactly to appending a regularizer to the attractive term of the loss, $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$. Beyond attracting pairs separated by large distances, the added term promotes an intermixing of points that facilitates convergence under random initialization, functioning much as early exaggeration does in t -SNE. We adopt the simplest linear correction in Eq. 13; alternative forms such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$) are equally admissible.

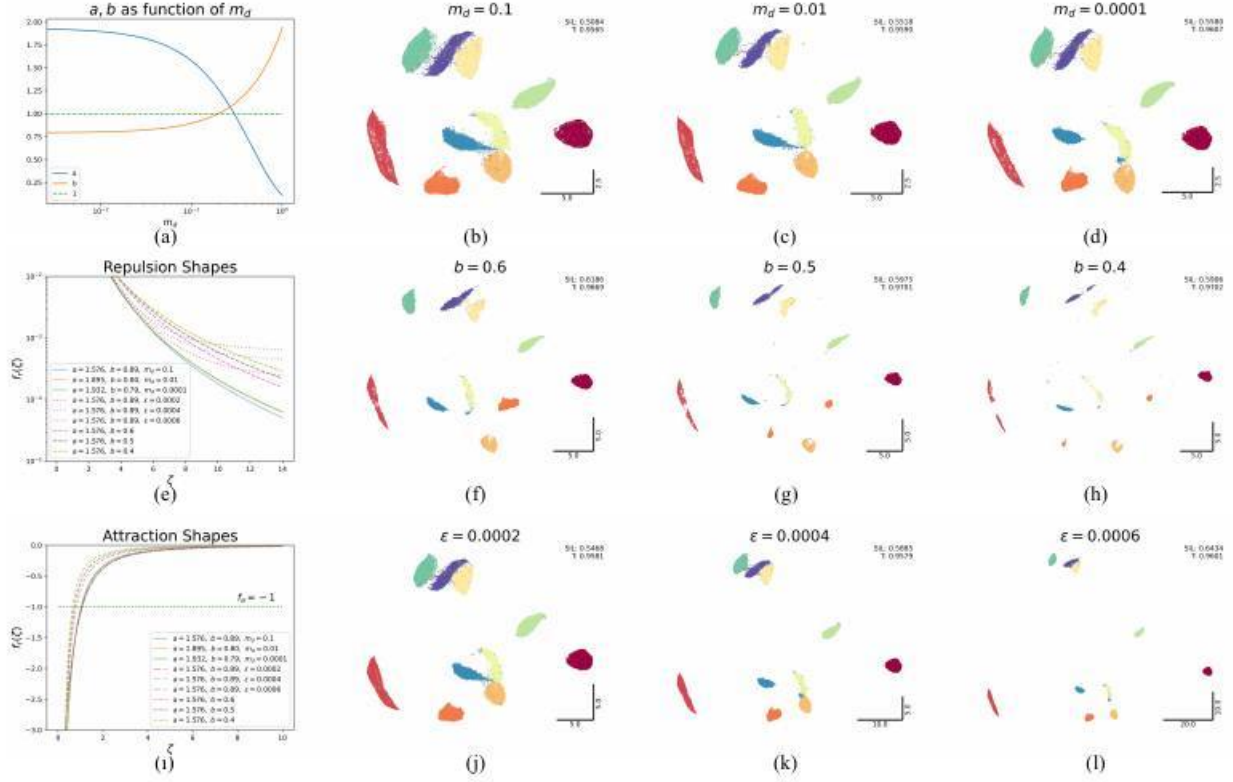


Figure 3: Control of inter-cluster distances on the MNIST dataset. (a) Deriving a, b by varying the low-dimensional distance m_d confines the reachable region of parameter space. (b–d) UMAP outputs for m_d set to 0.1, 0.01, and 0.0001, respectively, which yield negligible gains in cluster compactness. (e) Repulsion shapes across parameter settings. (f–h) Increasing repulsion by explicitly varying b produces markedly more compact clusters and resolves new clusters absent under the default. (i) Attraction shapes across parameter settings. (j–l) Increasing repulsion by adding a small positive constant ϵ to the repulsion shape enlarges inter-cluster distance while leaving attraction unchanged.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

By reverting to the original UMAP shape after the initial epochs, the composite form removes any distortion introduced by f_a^M once far-sighted intermixing has done its work. We now assess the effect of both the modified and the composite attraction shapes on dimensionality reduction from random initialization.

We first constructed a PCA-initialized embedding of the MNIST dataset (Fig. 2(a)) to serve as a reference, then generated embeddings from Gaussian random initialization for each attraction shape, repeating the procedure 100 times. To quantify consistency, we employ Procrustes analysis (Gower 1975), which aligns two point clouds under scaling, translation, rotation, and reflection (see Appendix D.1). Each randomly initialized embedding is aligned to the PCA reference, and its residual separation is measured by the Procrustes distance (p_d). Figure 2(b) presents the four UMAP embeddings with the lowest p_d : although the individual cluster shapes are preserved, their placements vary from run to run. The modified and composite attraction shapes (Figs. 2(c) and (d)) yield conspicuously more consistent cluster placements.

To characterize these placements more rigorously, we construct the Procrustes matrix, whose diagonal is sorted by p_d relative to the PCA-initialized mapping and whose off-diagonal entries record p_d between pairs of randomly initialized mappings (Appendix D.1); the construction is analogous to a similarity matrix (Foote, 1999). Embeddings produced by the default UMAP attraction shape are mutually dissimilar (Fig. 2(f)), whereas those produced by the modified (Fig. 2(g)) and composite (Fig. 2(h)) shapes are strongly similar to one another. This provides direct evidence that UMAP encodes a unique underlying structure irrespective of initialization; the default algorithm merely fails to attain that structure in the low dimension because its attraction decays too rapidly at long distances.

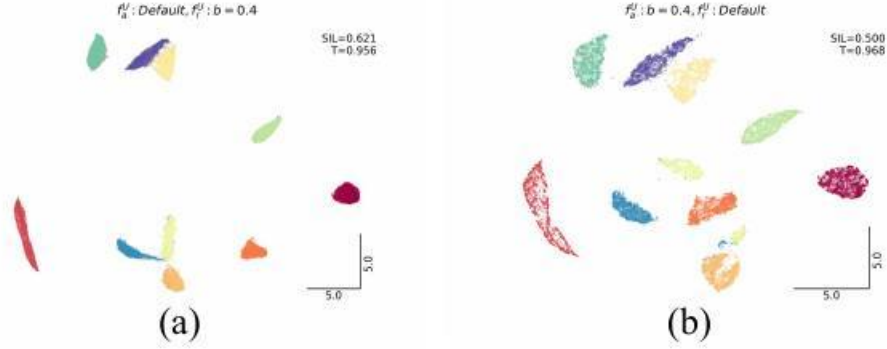


Figure 4: Embedding of the MNIST dataset under a controlled decoupling of the two forces: (a) default attraction shape combined with a repulsion shape set to $b = 0.4$, and (b) default repulsion shape combined with an attraction shape set to $b = 0.4$. In (a), increased repulsion reproduces the clusters of default UMAP with greater compactness; in (b), increased attraction develops new intra-cluster structure and resolves entirely new clusters. Attraction, not repulsion, is thus responsible for cluster formation.

5.2 Cluster Formation and Compactness

The parameter conventionally used to control cluster formation in UMAP is the minimum-distance parameter m_d (via Eq. 2). Varying m_d , however, confines the attainable values of a and b to a restricted locus (Fig. 3(a)), so that reducing m_d typically yields embeddings that confer no additional benefit (Figs. 3(b–d)). The reason is transparent in the shape formalism: varying m_d exerts only a marginal influence on the repulsion shape (Fig. 3(e)). A more direct route is to manipulate a and b explicitly. Decreasing a does raise repulsion, but it suppresses attraction even faster and thereby aggravates near-sightedness; decreasing b , by contrast, affords far finer control (Figs. 3(e,i)). Figures 3(f–h) show that lowering b to 0.6, 0.5, and 0.4 progressively enlarges inter-cluster distances and fragments existing clusters. This fragmentation follows from the increasingly heavy-tailed kernel induced by smaller b (heavy-tailed kernels yield smaller, more distinct clusters (Van der Maaten and Hinton, 2008, Kobak et al., 2019); see Kobak et al. (2019) for the precise dependence of the tail on b in UMAP). Although this manipulation cleanly separates all ten MNIST labels into distinct clusters, it does not by itself isolate the roles of the two forces: lowering b amplifies repulsion more strongly than lowering m_d , yet simultaneously reshapes the attraction profile—a confound we resolve next.

To isolate the effect of repulsion, we hold attraction fixed and perturb repulsion alone, adding a small positive constant ε to the repulsion shape,

$$f_r^M = f_r^U + \varepsilon, \quad (15)$$

holding a and b fixed. This perturbation is equivalent to adding a regularizer to the repulsive term of the loss, $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \varepsilon/2 \|y_i - y_j\|_2^2$, and delivers repulsion stronger than any attainable before (Fig. 3(e)). Figures 3(j–l) show that inter-cluster distances grow monotonically with ε . The clustering behavior, however, closely mirrors that obtained by varying m_d , producing only a loose separation of the labels as ε increases. Because ε leaves the attraction shape untouched and varying m_d traces essentially the same family of attraction shapes (Fig. 3(i)), these results converge on a single conclusion: cluster formation is governed predominantly by attraction.

To settle the question of cluster formation definitively, we modify the attraction shape and the repulsion shape one at a time—setting $b = 0.4$ in one while holding the other at its default—thereby decoupling the two forces that the previous manipulations had entangled (Fig. 4). With attraction held at its default, increased repulsion ($b = 0.4$) merely tightens the existing UMAP clusters and reveals no new structure (Fig. 4(a)). With repulsion held at its default, increased attraction ($b = 0.4$) instead resolves additional structure within each cluster and even splits some existing clusters into smaller ones (Fig. 4(b)). The dichotomy is unambiguous: attraction drives cluster formation, while repulsion controls cluster compactness—either by shrinking clusters or by enlarging inter-cluster distance, depending on the repulsion shape. Appendix G.3 shows that LocalMAP embeddings exploit precisely this attraction–repulsion interplay to separate clusters.

6 Comparison to NEG- t -SNE

Fixing $a = b = 1$ is a common convention across dimensionality reduction algorithms, as it renders the gradients in many methods stable (bounded) and thereby eases optimization. Damrich et al. (2023) recently pursued this route

for UMAP and proposed NEG- t -SNE, which not only stabilizes the gradient but also yields more compact clustering even at a constant learning rate of 1. Parametric UMAP (Sainburg et al., 2021), meanwhile, originally adopted the UMAP loss but subsequently switched to a numerically stable modified cross-entropy loss (Shi et al., 2023) built on a log-sigmoid kernel. Examined through the shape formalism, the two lines of work turn out to reach the same formulation from distinct starting points, which we state precisely as follows.

Proposition 6.1. *NEG- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

At $a = 1$ and $b = 1$, UMAP’s attraction and repulsion shapes reduce to $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. The attraction shape is now bounded within $[-2, 0]$, with $f_a^U(0) = -2$ (Fig. 1(f)), whereas the repulsion shape is essentially unaltered, still diverging as $\zeta \rightarrow 0$ (Fig. 1(g)). Because $f_a^U < -1$ as $\zeta \rightarrow 0$, Proposition 4.1 together with the analysis of Section 5 implies that even this unit-parameter case still requires learning-rate annealing.

Damrich et al. (2023) analyzed UMAP’s negative-sampling loss from a contrastive-embedding (CE) perspective and identified UMAP’s effective kernel as $1/\zeta^2$. Within the CE framework, they introduced NEG- t -SNE by replacing this kernel with $1/(1 + \zeta^2)$; translated back into the UMAP formalism, this corresponds to the low-dimensional affinity $q_{ij}^N = 1/(2 + \zeta^2)$, from which the attraction and repulsion shapes follow as $f_a^N = -2/(2 + \zeta^2)$ and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$ (Figs. 1(f,g)). The attraction shape now lies entirely within $[-1, 0]$ and thus satisfies Proposition 4.1 so every $\lambda \in [0, 1]$ yields contraction and suppresses the oscillation between expansion and contraction. NEG- t -SNE is correspondingly insensitive to learning-rate annealing, and its clusters remain sharp even at a constant $\lambda = 1.0$. Its repulsion shape, moreover, is bounded within $[0, 1]$ and stays finite as $\zeta \rightarrow 0$, so that far fewer points are ejected from clusters when sampled as negatives. Damrich et al. (2023) attributed this disparity solely to the contrast between NEG- t -SNE’s bounded repulsion and UMAP’s unbounded repulsion; our analysis in Fig. 1 shows that the attraction shape is equally responsible for cluster stability and compactness—an aspect their account overlooked. Appendix F provides further discussion and illustration. Read together with Fig. 4 this establishes that attraction and repulsion shapes drawn from different methods can be freely recombined—for instance, UMAP’s attraction shape with NEG- t -SNE’s repulsion shape—a possibility we demonstrate on UMAP, NEG- t -SNE, and PaCMAP in Appendix J.

7 Discussion and Conclusion

In this work, we established the relationship between the attractive and repulsive forces and cluster formation. Although it is a truism that attraction draws similar points together while repulsion drives dissimilar points apart, the precise mechanisms of these forces had not previously been characterized. We have demystified much of the dynamics that underlie cluster formation. While the main text concentrates on UMAP and NEG- t -SNE, the results are general to the broad family of attraction–repulsion methods, as the additional algorithms treated in Appendix C confirm.

Treating the attraction and repulsion coefficients as functional mappings and analyzing them through their respective shapes (Propositions 4.1 and 4.2) exposed a genuinely counterintuitive result: UMAP’s attractive force produces *repulsion*—expansion of distance—at short separations (Fig. 1(b)), pushing neighboring points apart rather than together. We conclude that UMAP’s learning-rate schedule exists precisely to suppress this effect.

We also examined the initialization of the low-dimensional embedding. Similar points that begin far apart experience attraction too weak to contract them adequately; our formalism supplies a principled means of strengthening attraction at large distances by augmenting the attraction shape. This modification produced embeddings markedly more consistent under random initialization, furnishing direct evidence that UMAP encodes a unique structure and that the customary numerical devices serve only to reach it more quickly.

Analysis of the repulsion shape established that stronger repulsion yields larger inter-cluster distances. Exploiting this, we added a small positive constant to the shape, achieving a level of repulsion unattainable by the conventional expedient of tuning the minimum-distance parameter m_d .

Finally, we clarified the relation between NEG- t -SNE and UMAP. In the case $a = 1$, $b = 1$, the essential distinction is that NEG- t -SNE, by default, has bounded attraction ($\in [-1, 0]$) and bounded repulsion ($\in [0, 1]$) shapes, so that both propositions hold irrespective of the learning rate; the method is therefore far less sensitive to any fixed $\lambda \in [0, 1]$. Once a and b are allowed to vary, however, NEG- t -SNE inherits the very numerical challenges of UMAP. We further proved that NEG- t -SNE and Parametric UMAP are equivalent.

These insights into attraction–repulsion dynamics constitute concrete tools for optimizing dimensionality reduction algorithms. Moreover, given the close correspondence between dimensionality reduction and contrastive learning (Damrich and Hamprecht 2021; Hu et al. 2023), our shape-based approach carries naturally to representation learning.

github.com, merged on April 26, 2022

Taken together, this work renders embeddings and their interpretation more principled, more consistent, and more reliable, and charts a clear course for future research.

Software and Data

All the data used in this paper are publicly available. The MNIST dataset is available at yann.lecun.com
Fashion-MNIST is available at github.com
Single-cell transcriptomes data is available at github.com

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Appendix

The appendices develop the analysis in full detail. Appendix A presents the derivations of UMAP’s attraction and repulsion shapes. Appendix B revisits the constant-learning-rate regime discussed in the main text. Appendix C demonstrates the benefits of improved attraction and repulsion on large language model embeddings. Appendix D.1 formally defines the Procrustes distance and matrix. Appendix E extends the random-initialization experiment to further datasets, and Appendix F extends the comparison of NEG- t -SNE and UMAP. Appendix G applies the shape formalism to a range of alternative dimensionality reduction methods, and Appendix H details, for completeness, the construction of the high-dimensional graph in each. Appendix I reports the detailed results of varying λ_a and λ_r (from Fig. 1) for UMAP, NEG- t -SNE, and PaCMAP across datasets, Appendix J demonstrates the effect of mixing and matching shapes across algorithms, and Appendix K records the implementation details.

A Proofs and Derivations

A.1 Proof of Proposition 4.1

Proof. We subtract Eq. (7) from Eq. (8) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

This distance contracts as long as $|1 + 2\lambda f_a| < 1$, i.e., provided

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition 4.2

Proof. We subtract Eq. (9) from Eq. (10) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

From the definition of f_r , the latter suffices. □

A.3 Attraction Shape:

We derive the attraction shape from a general low-dimensional affinity function, $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$, which recovers UMAP for $\gamma = 1$ and NEG- t -SNE for $\gamma = 2$, thereby treating both algorithms within a single derivation. The attractive force is given by

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

Defining $\zeta = \|y_i - y_j\|_2$, the first term gives the attraction shape as:

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

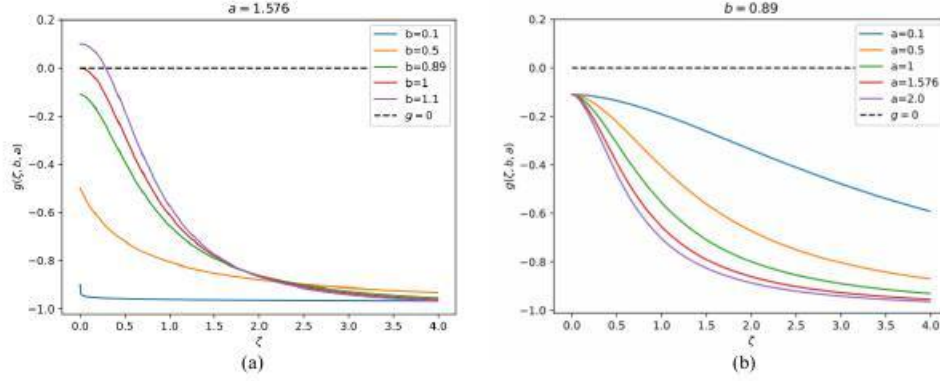


Figure 5: Values of $g(\zeta, b, a)$ (a) with a fixed at 1.576 and (b) with b fixed at 0.89. The shape f_a^U is strictly increasing precisely where $g < 0$, which holds throughout for $0 < b \leq 1$.

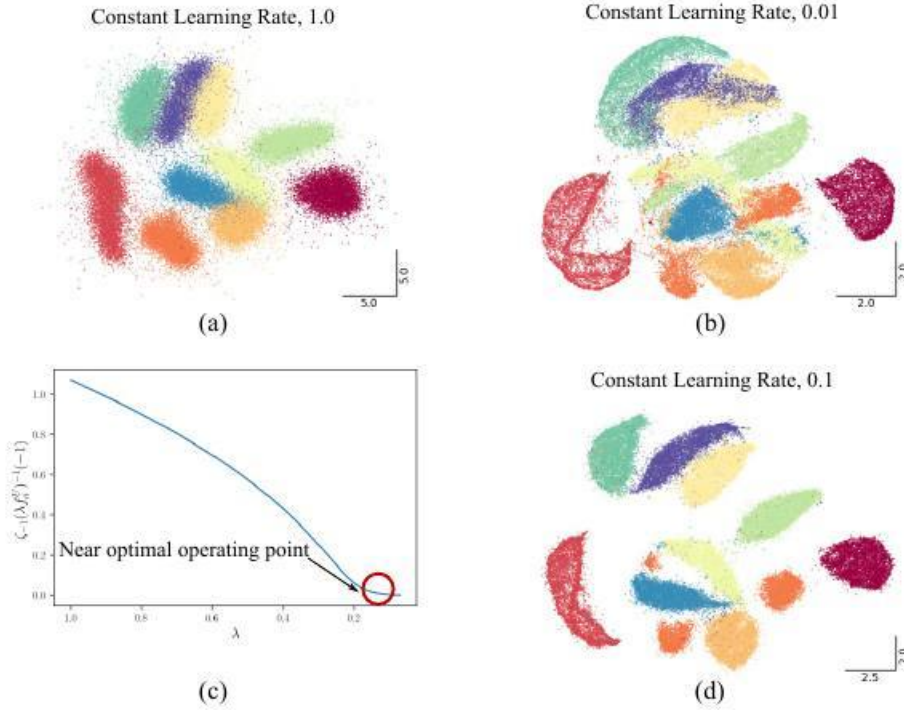


Figure 6: Effect of a constant learning rate on the embeddings. (a) When the learning rate is too high ($\lambda = 1.0$), the embedding is diffuse, owing to the large value of ζ_{-1} . (b) When the learning rate is too low ($\lambda = 0.01$), clusters fail to form, as both attraction and repulsion are too weak. (c) ζ_{-1} decreases nonlinearly with the learning rate; the objective is to shrink ζ_{-1} while retaining effective attraction and repulsion. (d) Distinct, compact clusters form at the near-optimal constant learning rate $\lambda = 0.1$.

A.4 Condition for strictly increasing f_a^U :

The behavior of f_a^U with distance is characterized by its derivative,

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

This leads to a strictly increasing condition ($\frac{df_a^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. This inequality holds for all $0 < b \leq 1$, as established by the derivative and asymptotes of g . Figure 5 plots g across b and a , confirming that the inequality (23) no longer holds once b exceeds 1.

A.5 Repulsion Shape:

Using the same general low-dimensional affinity, the repulsive force is given by

$$\begin{aligned} \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\ &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\ &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j). \end{aligned} \quad (24)$$

The first term gives the repulsion shape as:

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \quad (25)$$

Generally, $f_r > 0$.

A.6 Loss functions due to modified attraction and repulsion shape:

The cost function of the attractive term with the modification in Eq. (13) is given by

$$- \int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \quad (26)$$

whereas the repulsive term due to Eq. (15) is given by

$$- \int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \quad (27)$$

In both cases the additional term operates as a norm-based regularizer. The advantage of introducing it directly at the level of the attraction and repulsion shapes is interpretive: the role of each term becomes immediately legible—the attractive term increases far-sightedness, while the repulsive term contributes a constant repulsive coefficient.

B More on UMAP’s Learning Rate

As established in Sections 5 and 6 UMAP is widely held to require learning-rate annealing (Fig. 6). Our analysis in Section 5 explains why: we defined the minimum distance for contraction (ζ_{-1}) and showed that annealing the learning rate drives this quantity down and thereby produces compact clusters. In Section 6, comparing the attraction shapes of UMAP ($a = 1.0$, $b = 1.0$) and NEG- t -SNE, we explained that NEG- t -SNE tolerates a constant learning rate of 1.0 better than UMAP precisely because its attraction shape lies within $[-1, 0]$ whereas UMAP’s spans $[-2, 0]$. By the same reasoning, we showed that UMAP can withstand a constant learning rate of 0.5 once its attraction shape is confined to $[-1, 0]$ (Figs. 11(f,h)).

Even so, annealing yields better embeddings across a wide range of a and b , because the algorithm’s ultimate objective is to drive ζ_{-1} to zero, which occurs only as the learning rate itself approaches zero; otherwise the embedding remains diffuse (Fig. 6(a)). Conversely, an initial learning rate that is too low renders both attraction and repulsion too weak for any clear cluster to form (Fig. 6(b)). The ζ_{-1} -versus- λ curve (Fig. 6(c)) locates a near-optimal operating point at $\lambda = 0.1$, where ζ_{-1} is small yet attraction and repulsion retain appreciable strength; at this constant learning rate, we obtain compact clusters with clean boundaries (Fig. 6(d)).

C Large Language Model Embedding

The benefit of increased attraction and repulsion extends beyond image data to high-dimensional language model embeddings. To demonstrate this, we drew 300,000 labeled samples of PubMedBERT (Gu et al., 2021) features

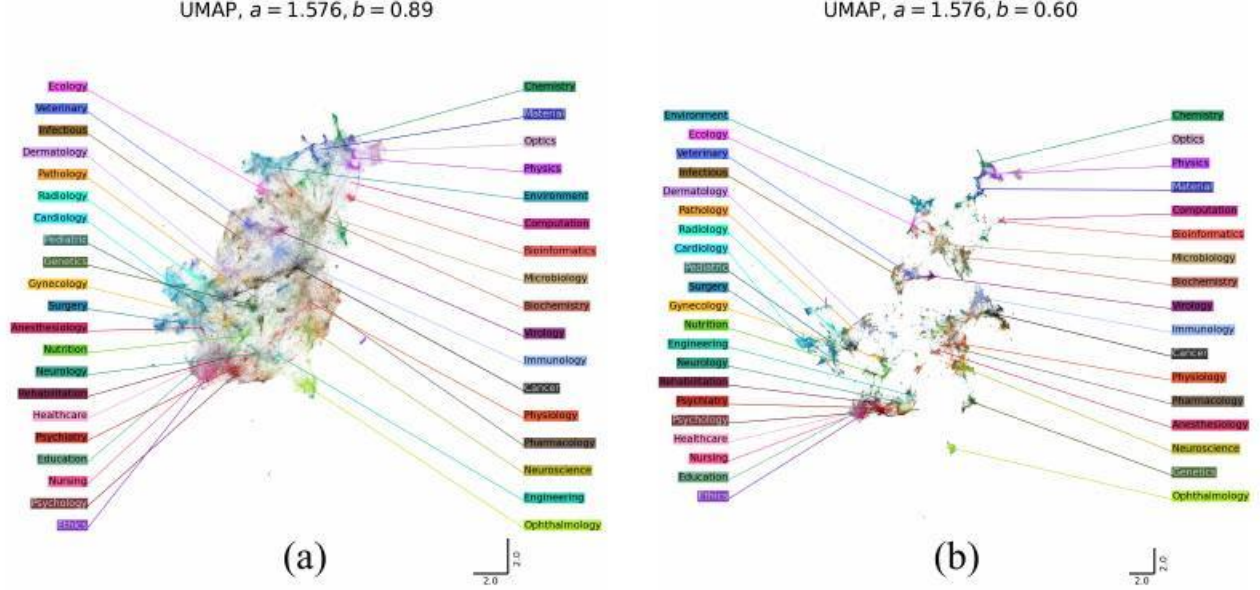


Figure 7: Visualization of 300K PubMed abstracts. (a) Default UMAP ($a = 1.576, b = 0.89$) produces heavily overlapping labels, whereas (b) UMAP with improved repulsion and attraction ($a = 1.576, b = 0.60$) yields well-formed, non-overlapping label clusters. Both mappings are labeled to aid interpretation.

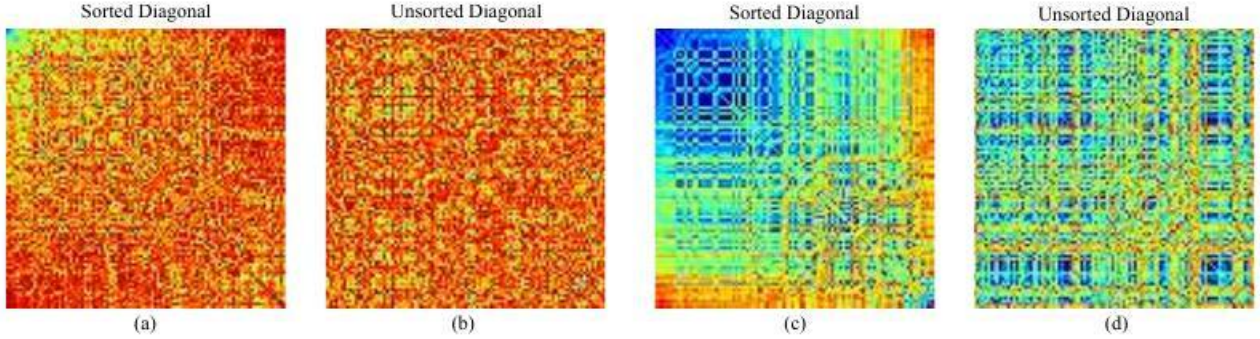


Figure 8: Effect of sorting the diagonal of the Procrustes matrix on its visualization. (a) Procrustes matrix reproduced from Fig. 2f for the default UMAP attraction shape, with the diagonal sorted by p_d relative to a PCA-initialized sample embedding. (b) The same data as (a) with an unsorted (randomly ordered) diagonal. (c) Procrustes matrix reproduced from Fig. 2g for the modified attraction shape, again with the diagonal sorted by p_d relative to a PCA-initialized sample embedding. (d) The same data as (c) with an unsorted diagonal. Sorting is purely a visualization aid: it exposes the self-similarity structure that an unsorted ordering conceals.

from the PubMed dataset (González-Márquez et al. 2024). Default UMAP produces a crowded structure in which the label colors overlap (Fig. 7a); absent the labels, the embedding would be difficult to interpret, as no explicit clusters emerge. Reducing b to 0.6—the mechanism analyzed in Section 5.2—simultaneously strengthens repulsion and, through improved attraction, promotes cluster formation, rendering the mapping readily explorable even without labels (Fig. 7b). This improvement is quantified by k -NN accuracy ($k = 10$): on a held-out set of 10,000 points, reducing b to 0.6 raises accuracy from 49.2% to 55.48%, a gain of more than six percentage points.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower, 1975) quantifies the similarity of two point clouds $\{x\}$ and $\{y\}$ under linear transformations—namely translation, scaling, and rotation. Operationally, we hold $\{x\}$ fixed and transform $\{y\}$

until the two sets attain maximum alignment. Letting $\{y'\}$ denote the optimally aligned transform of $\{y\}$, the Procrustes distance is

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

As a linear measure, the Procrustes distance has proven valuable across a range of settings (McInnes et al., 2018; Islam and Fleischer, 2022; Kotlov et al., 2024).

We use it here to measure the consistency of embeddings under random initialization. Let $\{x\}_p$ denote a reference embedding (obtained by PCA initialization in our experiments) and $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ a set of N embeddings obtained from random initialization. Their overall similarity is summarized by the mean and standard deviation of the strictly lower-triangular entries of the matrix P (reported as mean $\pm$ std in Figs. 2, 9 and 10).

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The indices of X_r may be ordered so that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$, whereupon the diagonal entries of the Procrustes matrix are

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal values are given by

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Numerically, sorting the diagonal changes nothing; visually, however, it exposes the underlying self-similarity of the embeddings. In Fig. 8(a), for instance, the sorted diagonal reveals that mutually similar embeddings cluster in the upper-left region of the matrix. Under the modified attraction shape, the count of mutually similar embeddings grows—evident in the enlarged blue region of Fig. 8(c)—signaling the presence of a metastable point in the embedding dynamics. When the diagonal is left unsorted, this region dissolves and all trace of self-similarity is lost (Fig. 8(b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski, 2001) quantifies the extent to which local neighborhoods are preserved under dimensionality reduction,

$$T = 1 - \frac{2}{nk(2n-3k-1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ denotes the k -NN graph in the embedding space and $r(i, j)$ the rank of x_j in the high-dimensional k -NN graph. We set $k = 5$, thereby assessing preservation of each point’s five nearest neighbors. For computational efficiency, we compute trustworthiness over 10,000 randomly sampled indices, holding the same indices fixed across all embeddings under comparison to ensure a like-for-like evaluation.

D.3 Silhouette Score

Although the silhouette score (Rousseeuw, 1987) was devised to assess clustering algorithms, we repurpose it to evaluate label separation in the embeddings—that is, how well the ground-truth labels are clustered—on the premise that an embedding algorithm naturally forms clusters and should separate labels as sharply as possible. For a point y_i in a point cloud $\{y\}$, let a_i denote the mean distance from y_i to the other points sharing its label and b_i the minimal mean distance from y_i to the points of any other label. The pointwise silhouette is then

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies in $[-1, 1]$: a value near 1 indicates that y_i sits comfortably within its own label cluster, a value near 0 marks a boundary point, and a value near -1 signals failed label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We computed the silhouette score over the entire embedding, without random sampling, using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

The main text (Section 5.1) reported results on MNIST alone. Here we replicate the experiment on Fashion-MNIST (FMNIST) (Xiao et al. 2017) and single-cell transcriptomes (Macosko et al. 2015) (Figs. 9 and 10 respectively). The conclusion is robust across all three datasets: modified and composite attraction shapes—those that strengthen attraction at large distances—substantially improve the consistency of reconstruction, as the reduced mean Procrustes distances in every panel attest.

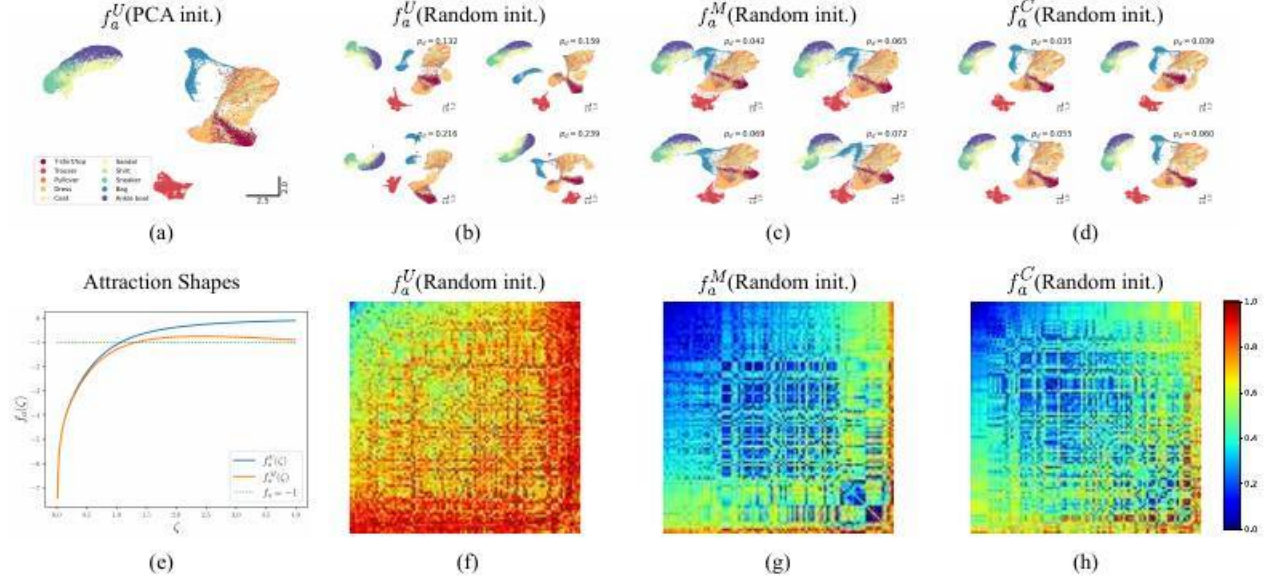


Figure 9: Effect of random UMAP initialization on different attraction shapes for the FMNIST dataset. (a) Reference mapping obtained with PCA initialization. (b–d) The four mappings with the lowest Procrustes distance (p_d) from the reference in (a), for the (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f–h) Procrustes matrices aggregated over 100 runs of the (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes. The modified and composite shapes roughly halve the mean p_d relative to the default.

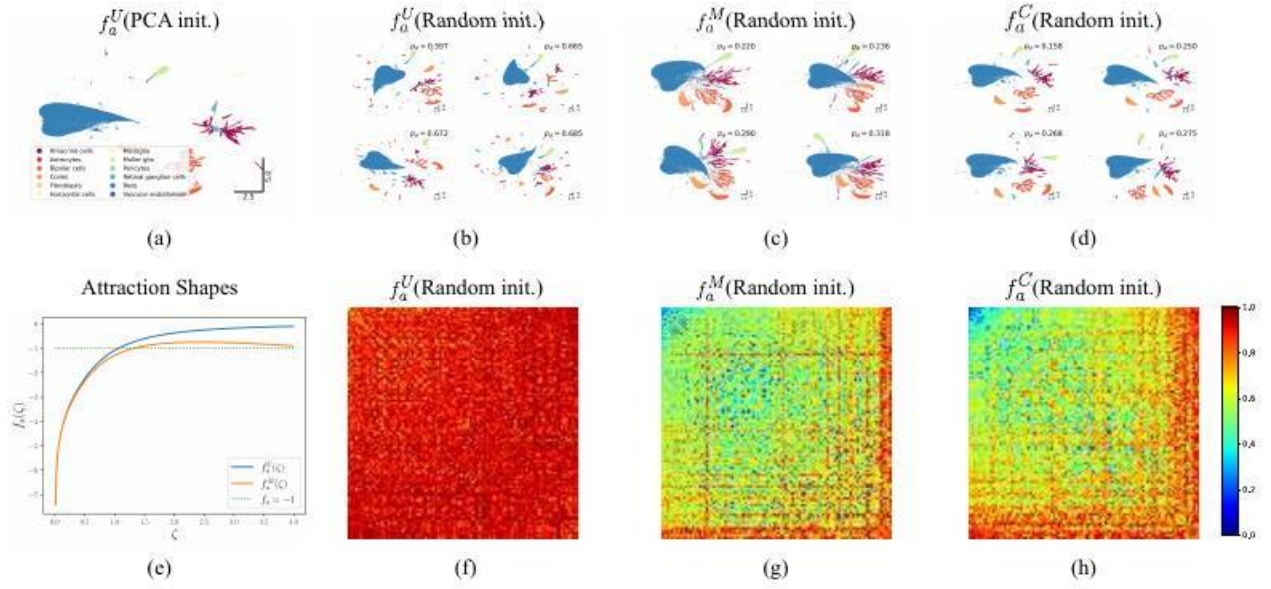


Figure 10: Effect of random UMAP initialization on different attraction shapes for the single-cell transcriptomes dataset. (a) Reference mapping obtained with PCA initialization. (b–d) The four mappings with the lowest Procrustes distance (p_d) from the reference in (a), for the (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f–h) Procrustes matrices aggregated over 100 runs of the (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes. Even on this challenging dataset, the modified and composite shapes markedly reduce the mean p_d .

F More on NEG- t -SNE

F.1 Comparison to UMAP

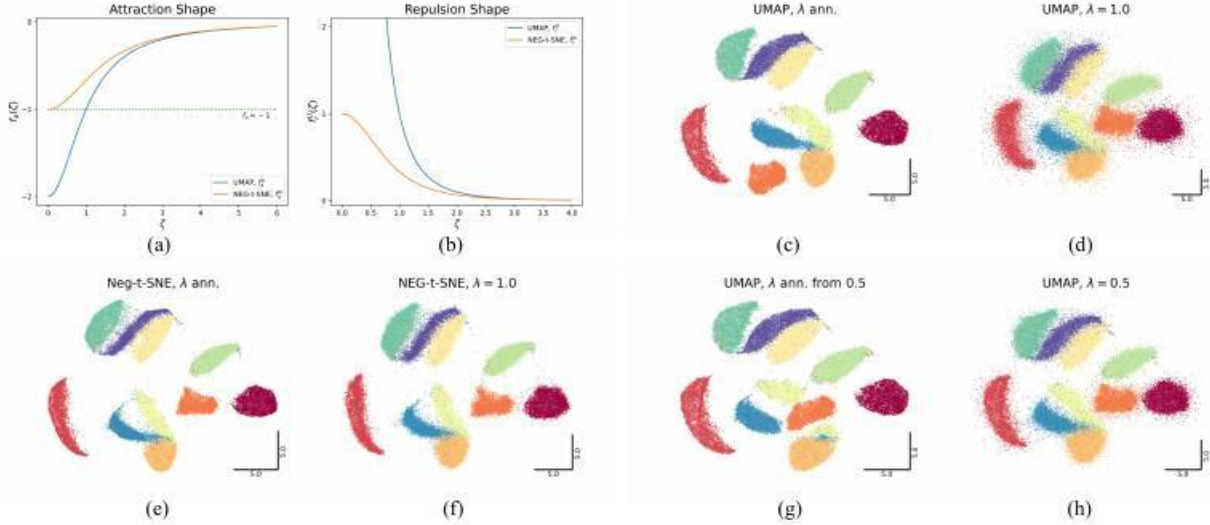


Figure 11: Sensitivity of UMAP and NEG- t -SNE to the learning rate on the MNIST dataset. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1$, $b = 1$) and NEG- t -SNE. (c,d) UMAP is highly sensitive to λ because $f_a^U < -1$ as the separation ζ decreases; without annealing, its clusters become fuzzy. (e,f) NEG- t -SNE is far less sensitive, since $f_a^N \in [-1, 0]$ throughout, so its clusters stay sharp even without annealing. (g,h) Confining f_a^U to $[-1, 0]$ by setting $\lambda = 0.5$ likewise reduces UMAP’s sensitivity to λ .

Figure 11 presents the shapes of UMAP and NEG- t -SNE alongside their MNIST embeddings. Under an annealed learning rate, the two methods produce comparable output (Figs. 11(c,e)); at a constant $\lambda = 1$, however, UMAP yields a fuzzy structure while NEG- t -SNE retains markedly cleaner boundaries (Fig. 11(d,f)). The analysis in Section 6 predicts that confining f_a^U to $[-1, 0]$ should sharpen UMAP’s clusters at fixed λ —a prediction already corroborated in Fig. 11(h,i), where UMAP achieved its best embedding and clustering for attraction learning rate $\lambda_a \lesssim 0.5$. The simplest way to enforce this is to set $\lambda = 0.5$, which satisfies Proposition 4.1 for all ζ . The resulting embeddings (Figs. 11(g) and (h)) confirm the prediction: the clusters now resemble those of NEG- t -SNE and, at constant $\lambda = 0.5$, exhibit sharper boundaries than before (Fig. 11(d,h)). A few points remain outside the clusters, a residual artifact of UMAP’s repulsion shape that NEG- t -SNE resolves.

We may next introduce the parameters a and b into NEG- t -SNE—which is precisely the formulation of Parametric UMAP (Proposition 6.1). The affinity function becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes become $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$ and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes diverge as $\zeta \rightarrow 0$, so NEG- t -SNE incurs the same numerical challenges as UMAP once a and b are varied, and the corresponding limitations carry over intact. One notable distinction is that its attraction shape attains a smaller minimum distance for contraction ζ_{-1} than UMAP’s; although this can aid cluster formation, the shape also approaches zero more rapidly—an intensified near-sightedness at large distances—which may curtail its effectiveness for attraction over longer separations.

F.2 Comparison to Parametric UMAP

Parametric UMAP was originally trained with the UMAP objective (Sainburg et al., 2021), but subsequent work adopted a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al., 2023). This modification renders Parametric UMAP and NEG- t -SNE (Damrich et al., 2023) equivalent (Proposition 6.1), as we now prove.

Proof of Proposition 6.1

Proof. The kernel function under this modification becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both expressions coincide with NEG- t -SNE carrying explicit parameters a and b (which NEG- t -SNE fixes to 1). $\square$

G Alternate Dimensionality Reduction Algorithms

The alternative algorithms we analyze all employ the same low-dimensional kernel as UMAP (with $a = 1$ and $b = 1$ in their low-dimensional weight),

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

We begin with TriMap (Amid and Warmuth, 2019). Although TriMap operates on triplets rather than pairwise interactions, it serves as an instructive primer for attraction and repulsion analyses somewhat more involved than UMAP's. We then treat Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al., 2021) and its extension, Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al., 2025), both of which adapt TriMap's loss to pairwise interactions. Next we address t -SNE (Van der Maaten and Hinton, 2008), followed by a brief note on SNE (Hinton and Roweis, 2002), which uses a different kernel, and we close with multidimensional scaling (Borg and Groenen, 2007). In every case, the shape formalism developed for UMAP applies with only minor extension, underscoring its generality.

G.1 TriMap

TriMap (Amid and Warmuth, 2019) optimizes the low-dimensional embedding through a triplet loss,

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} is the weight of the triplet (y_i, y_j, y_k) , y_j belongs to the high-dimensional k -nearest-neighbor set of y_i , and y_k is a distant point. Minimizing this loss draws y_i and y_j together while pushing y_i and y_k apart. The corresponding update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the distance between nearest neighbors, $\zeta_2 = \|y_i - y_k\|_2$ is the distance from the faraway point, and f_a^T and f_r^T are attraction and repulsion shapes of TriMap, respectively. Unlike UMAP, the attractive and repulsive components are non-separable and the shapes depend on two distance measures (making them 2D). The functional form of the attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

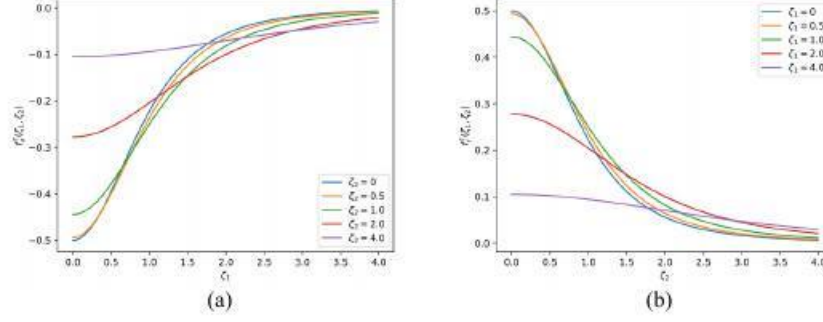


Figure 12: (a) Attraction shapes of TriMap across values of ζ_2 and (b) repulsion shapes of TriMap across values of ζ_1 , both revealing the two-distance dependence absent in UMAP.

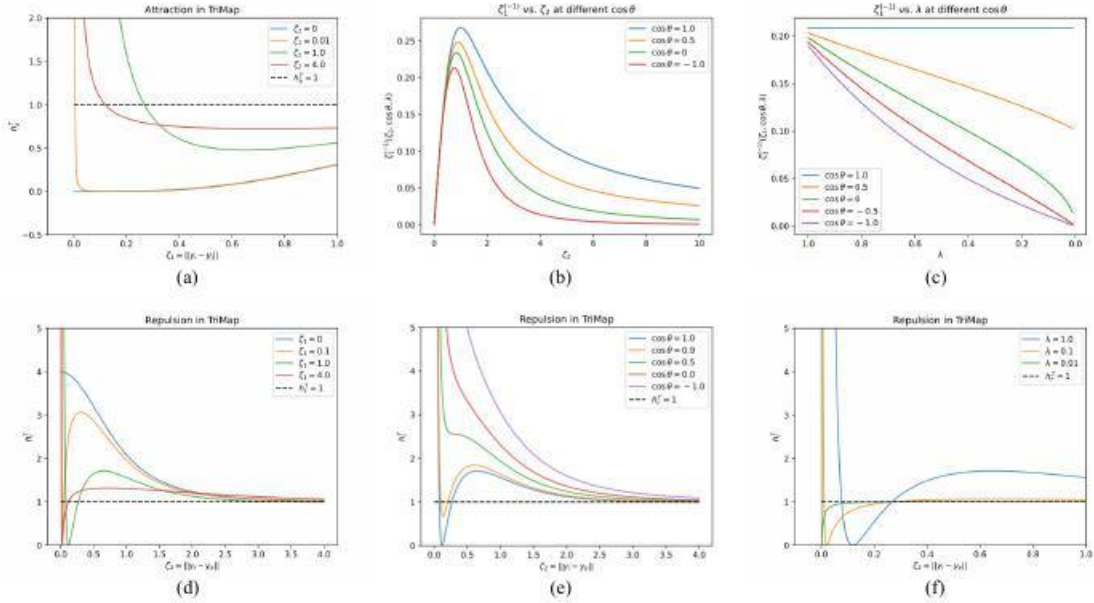


Figure 13: Attraction and repulsion behavior in TriMap. (a) h_a^T versus ζ_1 across values of ζ_2 ; values below the dotted line indicate attraction. As in UMAP, TriMap exhibits both attraction and repulsion for the nearest-neighbor pair (y_i, y_j) . (b,c) Unlike UMAP, the minimum distance for contraction $\zeta_1^{(-1)}$ varies with (b) ζ_2 and (c) λ , with $\cos \theta$ setting the range of these values. (d-f) Repulsion in TriMap as (d) ζ_1 , (e) $\cos \theta$, and (f) λ vary; values above (below) the dotted line indicate repulsion (attraction). Whereas UMAP's repulsive force yields only repulsion, TriMap's can yield both attraction and repulsion. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

TriMap’s attraction and repulsion shapes (Fig. 12) follow trends broadly similar to UMAP’s. A key difference is that the attraction shape bottoms out at -0.5 , so that—unlike UMAP—attraction alone never flips the relative positions of a pair. Because Eqs. (45)–(47) do not decouple the attractive and repulsive terms, however, Propositions 4.1 and 4.2 no longer apply directly. Treating attraction first, we establish the following.

Proposition G.1. *Update equations (45)–(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i - y_j)^T (y_i - y_k)}{\|y_i - y_j\|_2 \|y_i - y_k\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends jointly on ζ_1 , ζ_2 , $\cos \theta$, and λ ; in particular, the distance $\zeta_1 = \|y_i - y_j\|_2$ that we wish to contract is coupled to these additional variables. Figure 13(a) traces the attraction behavior across values of ζ_2 at $\cos \theta = 1$ and $\lambda = 1$: values below the dotted line denote attraction (contraction of ζ_1) and values above it denote repulsion (expansion of ζ_1). The intersection of the dotted line with h_a^T marks the minimum distance for contraction $\zeta_1^{(-1)}$ —the TriMap analogue of UMAP’s ζ_{-1} —which we define as

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

$\zeta_1^{(-1)}$ is finite and strictly positive in most cases (Fig. 13(b,c)). Consequently, TriMap can exhibit the same contraction–expansion behavior as UMAP and therefore requires learning-rate annealing or an equivalent device—in the reference implementation, the delta-bar-delta (Jacobs, 1988) method under suitable initialization.

Proposition G.2. *Update equations (45)–(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

Inequality (54) likewise depends on ζ_1 , ζ_2 , $\cos \theta$, and λ ; here the quantity of interest is the expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d–f) trace the repulsion behavior as the remaining quantities vary, with values above the dotted line denoting repulsion (expansion) and values below it denoting attraction. The striking contrast with UMAP is that TriMap’s repulsion can produce contraction rather than expansion; because this anomaly is confined to small distances, however, it can be averted by appropriate initialization and triplet selection.

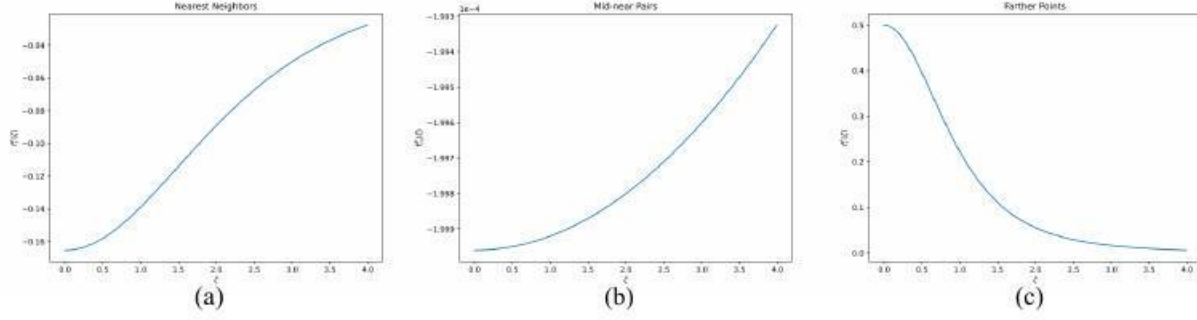


Figure 14: Attraction and repulsion shapes of PaCMAP. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near pairs (note that the latter are on the order of 10^{-4}). (c) Repulsion shape for further pairs. In all panels, $\lambda = 1$.

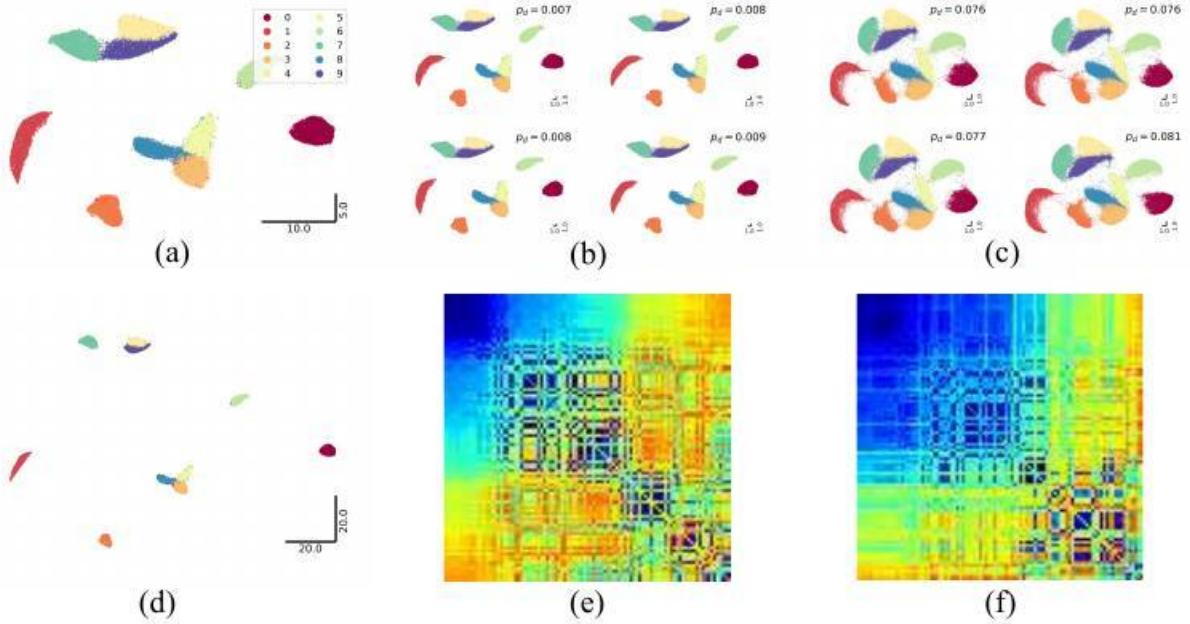


Figure 15: PaCMAP behavior under different conditions. (a) PaCMAP of MNIST with PCA initialization. (b,c) The four randomly initialized embeddings best matching (a) for (b) original PaCMAP and (c) PaCMAP with the modified attraction shape $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$. (d) PaCMAP of MNIST with the modified repulsion shape $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) original PaCMAP (0.51 ± 0.22) and (f) PaCMAP with the modified attraction shape (0.43 ± 0.22). Exactly as for UMAP, increased attraction at large distances improves consistency, while increased repulsion yields smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) optimizes the low-dimensional embedding across multiple scales simultaneously. Its loss function is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} weight the nearest-neighbor (NN) pairs, mid-near (MN) pairs, and further pairs (FP), respectively (see Appendix H). The first two terms furnish attraction and the last furnishes repulsion. The loss is, in effect, a modified TriMap triplet loss: for the attractive terms it substitutes constant coefficients for TriMap’s distant-point affinity (1/10 for nearest neighbors, 1/10000 for mid-near pairs, and 1 for further points). It is thus

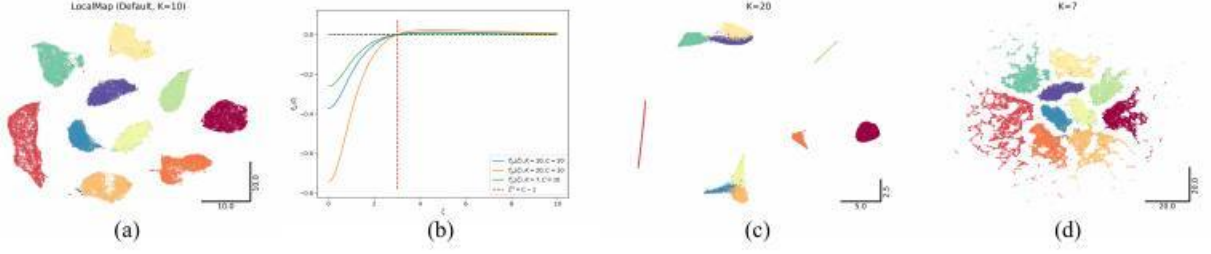


Figure 16: Behavior of LocalMAP on MNIST. (a) Default embedding. (b) Attraction–repulsion shape for nearest neighbors; values below (above) the dotted line indicate attraction (repulsion), with the transition from attraction to repulsion occurring as ζ crosses $\sqrt{C-1}$. Following the LocalMAP implementation, we set $C = 10$. (c) For large K ($= 20$), repulsion dominates and clusters become compact. (d) For small K ($= 7$), attraction dominates and clusters break up.

separable into three terms, which decouples the update equations. For the nearest-neighbor term, the updates are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

of the mid-near pairs are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and of the farthest pairs are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, and $\zeta_3 = \|y_i - y_l\|_2$ are the respective distances, f_a^P and f_m^P are the attraction shapes for the nearest-neighbor and mid-near pairs, and f_r^P is the repulsion shape for the further pairs. The corresponding functional forms are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

Here f_a^P and f_m^P obey Proposition 4.1 and f_r^P obeys Proposition 4.2 (Fig. 14). PaCMAP’s attraction is much weaker than UMAP’s, yet it suffices across a broad range of learning rates (modulated by Adam (Kingma and Ba 2015)); with $w_{NB} = 3$, the maximum attraction stays below 0.5, precluding any position flips during the attractive update. PaCMAP typically initializes the embedding within a small sphere in the low dimension—on the order of 10^{-3} in (Wang et al. 2021)—and relies on repulsion to separate the individual clusters. It thereby recovers most of UMAP’s clustering properties (particularly on MNIST) with improved global ordering, and its consistency under random initialization surpasses UMAP’s (Fig. 15(b,e))—a margin our modified attraction widens still further (Fig. 15(c,f)). As with UMAP, adding a small constant to the repulsion shape strengthens repulsion and thus increases embedding compactness by enlarging inter-cluster distance.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al. 2025) is a refinement of PaCMAP whose defining achievement is the clean separation of all ten MNIST clusters (Fig. 16(a)), behavior that mirrors Figs. 3(g,h,j,k) and 4(a,b). We analyze how the interplay of attractive and repulsive forces governs cluster compactness and connectedness.

LocalMAP first runs PaCMAP and then performs additional optimization on the attraction in order to decouple certain clusters, minimizing the loss

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term fuses the attractive and repulsive character of the triplet loss into a single expression acting on the same pair: it induces attraction in one regime and repulsion in the other. The second term is identical to PaCMAP's, save that further pairs are resampled every few iterations. We therefore analyze only the first, nearest-neighbor term, whose update equations are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_j^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction-repulsion shape given by

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

By Proposition 4.1 the update contracts whenever $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$; when $\zeta^2 > C-1$ we have $f_{ar}^L > 0$, and by Proposition 4.2 the update expands. The parameters K and C thus determine whether attraction or repulsion dominates the dynamics. The LocalMAP implementation fixes $C = K = 10$ (Fig. 16(b)), which makes the attraction stronger than PaCMAP's (Fig. 14(a)). Consequently, for $\zeta > 3$ the nearest neighbors experience repulsion, so that pairs bridging two clusters are driven apart, with K acting as an overall force-scaling parameter that regulates this separation.

By Proposition 4.1 the condition $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ bounds the largest admissible K —and hence the maximum attraction that f_{ar}^L can supply—without flipping the placement of a pair. (Avoiding such flips during LocalMAP optimization is essential to preserve the ordering inherited from PaCMAP, which would otherwise impede cluster separation.) This bound simplifies Eq. (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, giving $K \lesssim 13.44$ for $C = 10$ and $\lambda = 1$. For $K \gtrsim 13.44$, repulsion dominates and the clusters grow more compact, yet the LocalMAP objective fails—the bridge between clusters persists (Fig. 16(c), $K = 20$). Conversely, as K decreases, attraction dominates (repulsion being too weak) and existing clusters break apart (Fig. 16(d), $K = 7$), reproducing the pattern of Fig. 4(b). These two limits confirm, within LocalMAP, the same attraction-forms-clusters/repulsion-compacts-clusters division established for UMAP.

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton [2008]) optimizes pairwise similarities. Its loss function is

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the weight of the pair and the original implementation considers all pairs, not merely nearest neighbors. This loss decomposes into attractive and repulsive contributions as

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term supplies the attractive forces and the second the repulsive forces. The attractive term is identical to UMAP's (at $a = 1$, $b = 1$) and is inexpensive to evaluate, whereas the repulsive term couples every pair through the normalizer and is correspondingly costly. Applying the same decomposition principles we used for UMAP yields the t -SNE update equations. Because the original t -SNE does not restrict to a nearest-neighbor graph, the weight $w_{i,j}$ —computed over all pairs—enters these updates explicitly. The attractive updates are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the attraction shape of t -SNE and $\zeta_{i,j} = \|y_i - y_j\|_2$. The update equation for the repulsive parts are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k})(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j})(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the repulsion shape of t -SNE and $Z = \sum_{k \neq l} q_{k,l}$ is the normalizer. The functional forms of the two shapes are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1 + \zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1 + \zeta^2)^2}. \quad (77)$$

From the attractive updates (72)–(73), f_a^{t-SNE} obeys Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$) and admits a well-defined minimum distance for contraction, ζ_{-1} . The repulsive update, coupled across all pairs, bears no simple relation to the repulsion shape; leveraging the analysis developed for TriMap, however, we can derive the following.

Proposition G.3. *The update Eqs. (74)–(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k, k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l, l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. (79), we obtain the desired inequality. $\square$

Because t -SNE’s repulsion condition (Eq. 78) mirrors TriMap’s, the two share the same repulsion behavior; in particular, t -SNE’s repulsive forces can yield both attraction and repulsion.

Since t -SNE’s forces are scaled—by $w_{i,j}$ for attraction and by $\frac{w_{i,j}}{Z}$ for repulsion—both are typically weaker than UMAP’s, and the algorithm accordingly employs large learning rates (e.g., $\approx 10^3$ in the Open- t -SNE package (Poličar et al. [2024])). Most t -SNE implementations further require an “early exaggeration” step, in which the attractive forces are multiplied by a constant during the first few iterations. This device pulls together points that belong close but are initially placed far apart—inducing far-sightedness—and, symmetrically, separates points that are very close ($\zeta_{i,j} \approx \zeta_{-1}$) yet should be apart. Early exaggeration is therefore indispensable for obtaining a consistent t -SNE embedding, particularly under random initialization; applied throughout the optimization, it even draws t -SNE embeddings toward the appearance of UMAP embeddings (Böhm et al. [2022]).

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis [2002]), among the oldest algorithms in this class, shares t -SNE’s loss (Eq. 71) but adopts the Gaussian low-dimensional kernel $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. This choice produces the constant attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which obeys Proposition G.3 after substituting the SNE counterparts for the t -SNE symbols. Being constant, the attraction shape is ill-posed with respect to distance: contraction depends entirely on the learning rate λ and the weights $w_{i,j}$, so that clusters overlap even for small sample sizes—the crowding problem (Van der Maaten and Hinton [2008]). t -SNE and its successors remedy this precisely by abandoning the Gaussian kernel in favor of the heavy-tailed kernel of Eq. (43), a transition our shape analysis renders explicit.

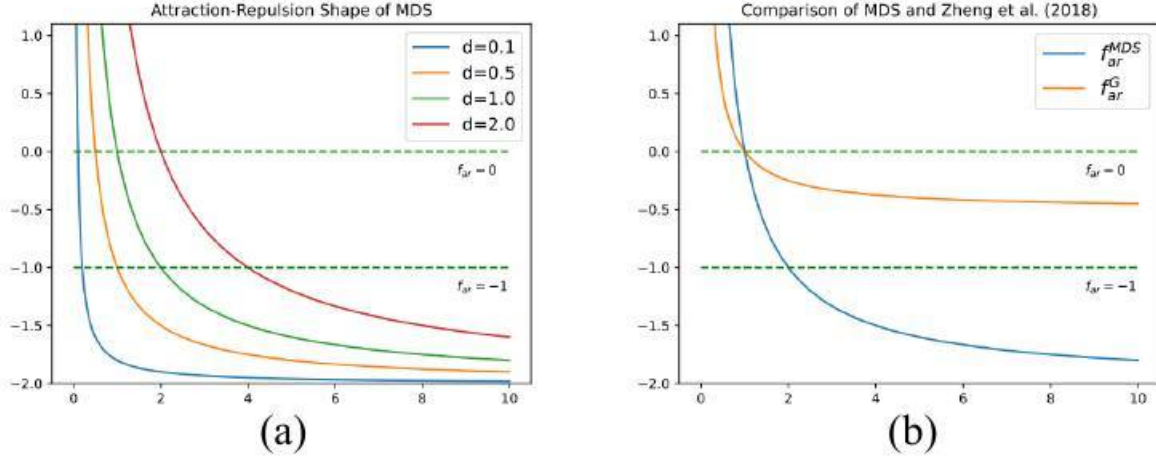


Figure 17: (a) Attraction–repulsion characteristic of MDS expressed through f_{ar} . (b) Comparison of the attraction–repulsion characteristics of MDS and the method of Zheng et al. (2018) at $d = 1.0$.

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen, 2007) conventionally eschews gradient methods, which often fail to converge to good mappings, in favor of stress majorization; nonetheless, a few works pursue gradient-based formulations (Kruskal, 1964; Zheng et al., 2018), and we treat this case briefly here. In particular, Zheng et al. (2018) develops a successful gradient-descent MDS algorithm for graph drawing. We begin from the cost function (the weight $w_{i,j}$ may be absorbed into the learning rate without loss of generality),

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss contains a single term per pair, with no explicit separation into attractive and repulsive parts; that lone term supplies both attraction and repulsion. Fortunately, the loss is separable across the (i, j) pairs, so a single pair may be analyzed in isolation. Its derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

Thus, the attraction-repulsion shape of MDS becomes

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. The shape is attractive when $\zeta > d$, in which regime it is bounded within $[-2, 0]$ (Fig. 17(a)). This range is nonetheless unsuited to convergence: only values within $[-1, 0]$ contract while the remainder expand, and values below -2 flip the points and expand them further. To circumvent this, Zheng et al. (2018) introduces an ad hoc formulation inspired by force-directed graph layout,

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which is bounded within $[-0.5, 0]$ and hence remains valid for any effective learning rate $\lambda \leq 1$. For $\zeta < d$, by contrast, $f_{ar}^{MDS} > 0$ and the interaction becomes repulsive. This attraction–repulsion interaction performs best when the initialization already lies close to the target output; under continued optimization, a pair oscillates about the distance $\zeta = d$ where $f_{ar}^G = 0$, furnishing a notion of transition distance analogous to UMAP’s ζ_{-1} . Crucially, no learning rate drives this distance to zero—indeed, maintaining it is the very objective of MDS—so MDS clusters tend to be fuzzier than those of UMAP or PaCMAP (see Lambert et al. (2022), de Bodt et al. (2025), and Kury et al. (2025) for illustrations). We note that Lambert et al. (2022) recasts the MDS loss as a quartet stress (over four samples) with a relative distance (normalized by the sum of the six quartet distances), enabling global regularization at lower computational cost than the majorization approach.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) [\[Hinton and Roweis, 2002\]](#) underpins modern dimensionality reduction algorithms. It builds a high-dimensional graph of the dataset $X = \{x_i \in R^n | i = 1, \dots, N\}$ through the system

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is set so as to match a user-specified *perplexity* P , defined by

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP, by contrast, constructs its high-dimensional graph via the k -nearest-neighbor (k -NN) algorithm through the system

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ denotes the set of k -nearest neighbors of x_i and σ_i is a scaling parameter chosen so that $\sum_j p_{i|j} = \log_2(k)$. The graph is subsequently symmetrized by a t-conorm,

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses just the k -NN graph for the affinities (all equal to 1) with a self-tuning distance measure [\[Zelnik-Manor and Perona, 2004\]](#),

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average distance from x_i to its fourth through sixth Euclidean nearest neighbors. This σ_i serves the same purpose as its counterparts in Eqs. [\(87\)](#) and [\(90\)](#), notwithstanding their differing definitions.

Whatever these graph-construction choices, our analysis shows that it is the optimization—and, within it, the attraction and repulsion shapes—that ultimately drives the low-dimensional embedding.

I Detailed Results for Varying λ_a and λ_b

This section reports the detailed results underlying the experiments summarized in Fig. 11(h,i), obtained by substituting the attraction and repulsion shapes of the other methods into UMAP. Figure 18 reproduces the main-text results. Figures 19–22 display the individual embeddings for each choice of λ_a and λ_r per method, together with the initialization, a reference embedding under annealed λ , and the degenerate cases in which λ_a or λ_r is set to 0. For PaCMAP, which additionally employs mid-near pairs, we include a further reference incorporating those pairs. Figures 23–27 and 28–32 present the corresponding results on FMNIST and single-cell transcriptomes, respectively.

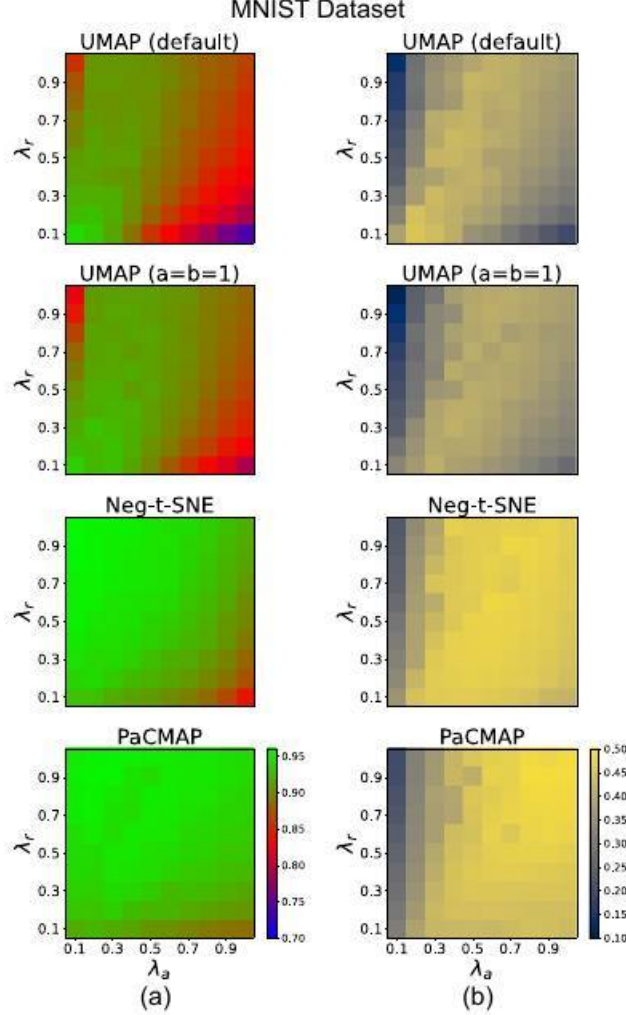


Figure 18: (a) Trustworthiness and (b) Silhouette score across methods for the MNIST dataset, as λ_a and λ_r are varied independently.

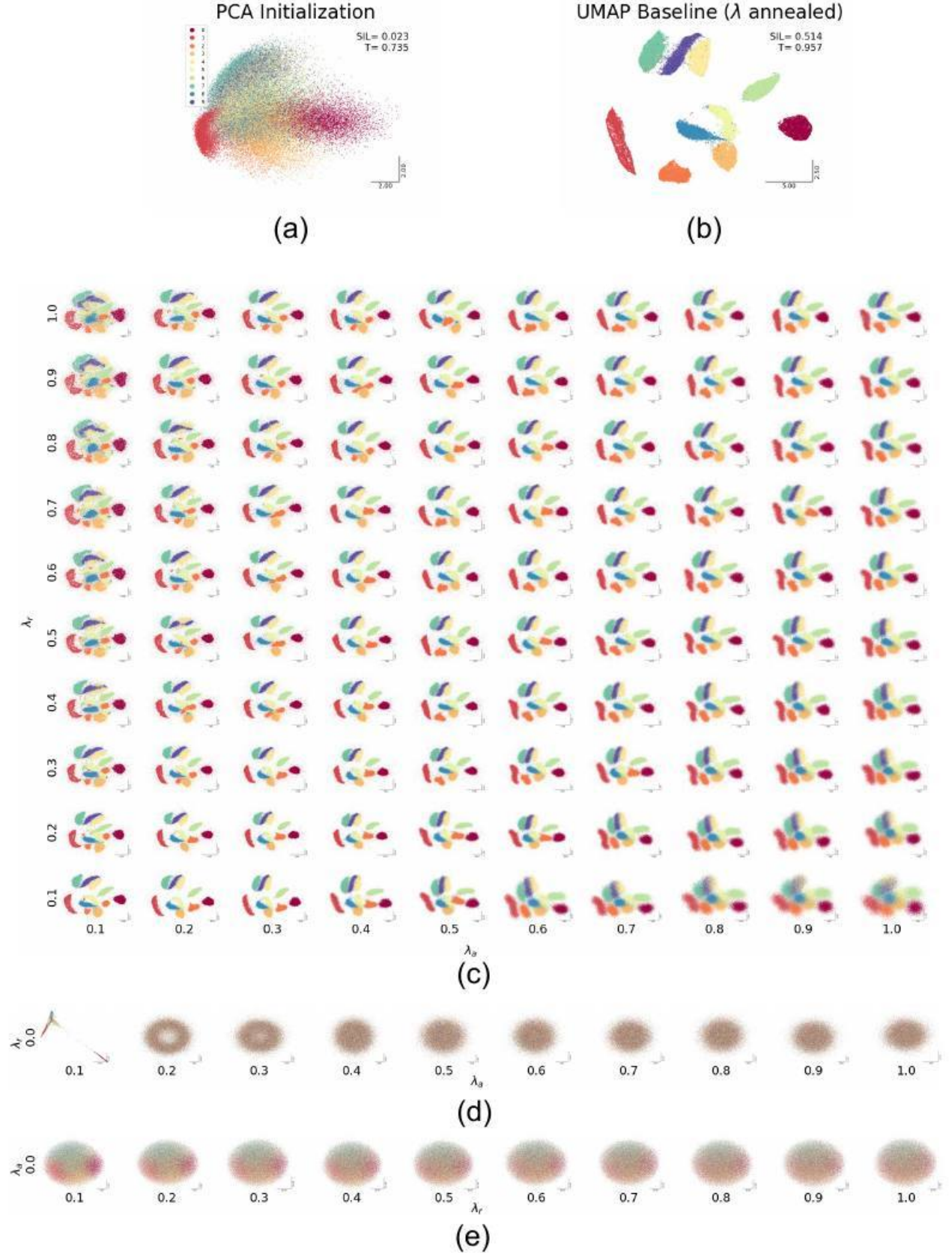


Figure 19: Varying λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline with λ annealed from 1. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

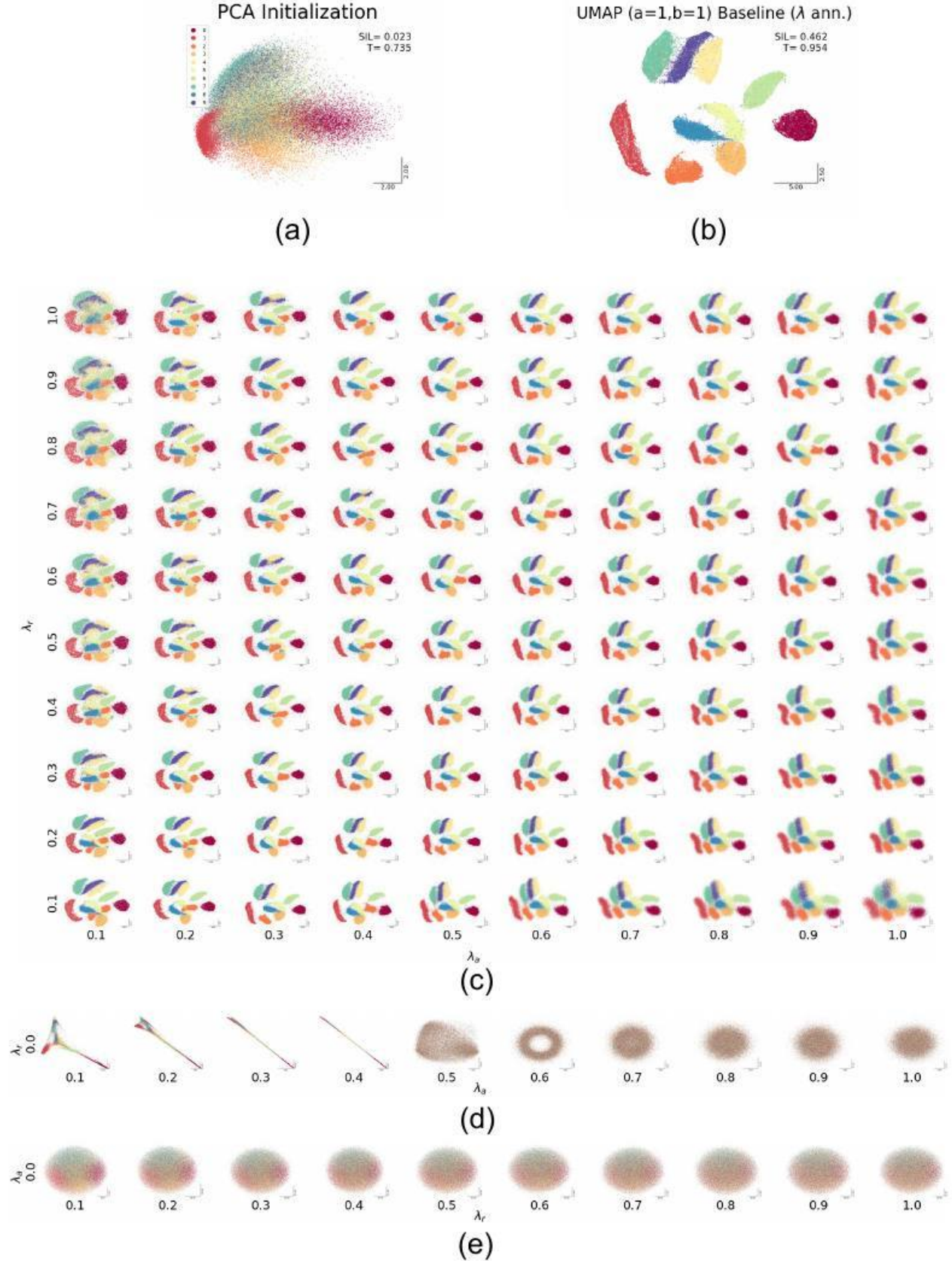


Figure 20: Varying λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization of the embeddings. (b) Baseline with λ annealed from 1. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

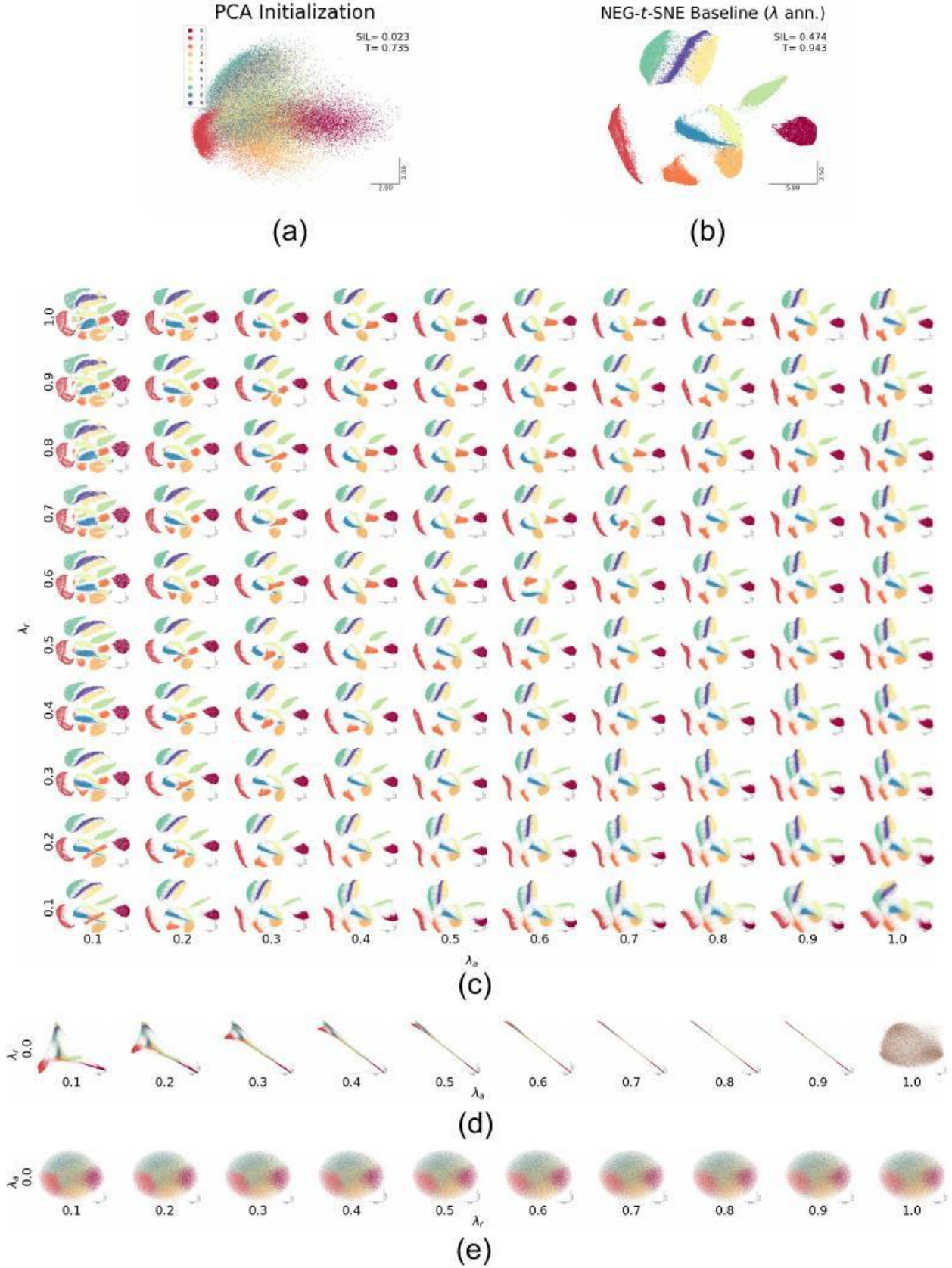


Figure 21: Varying λ_a and λ_r for the MNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline with λ annealed from 1. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

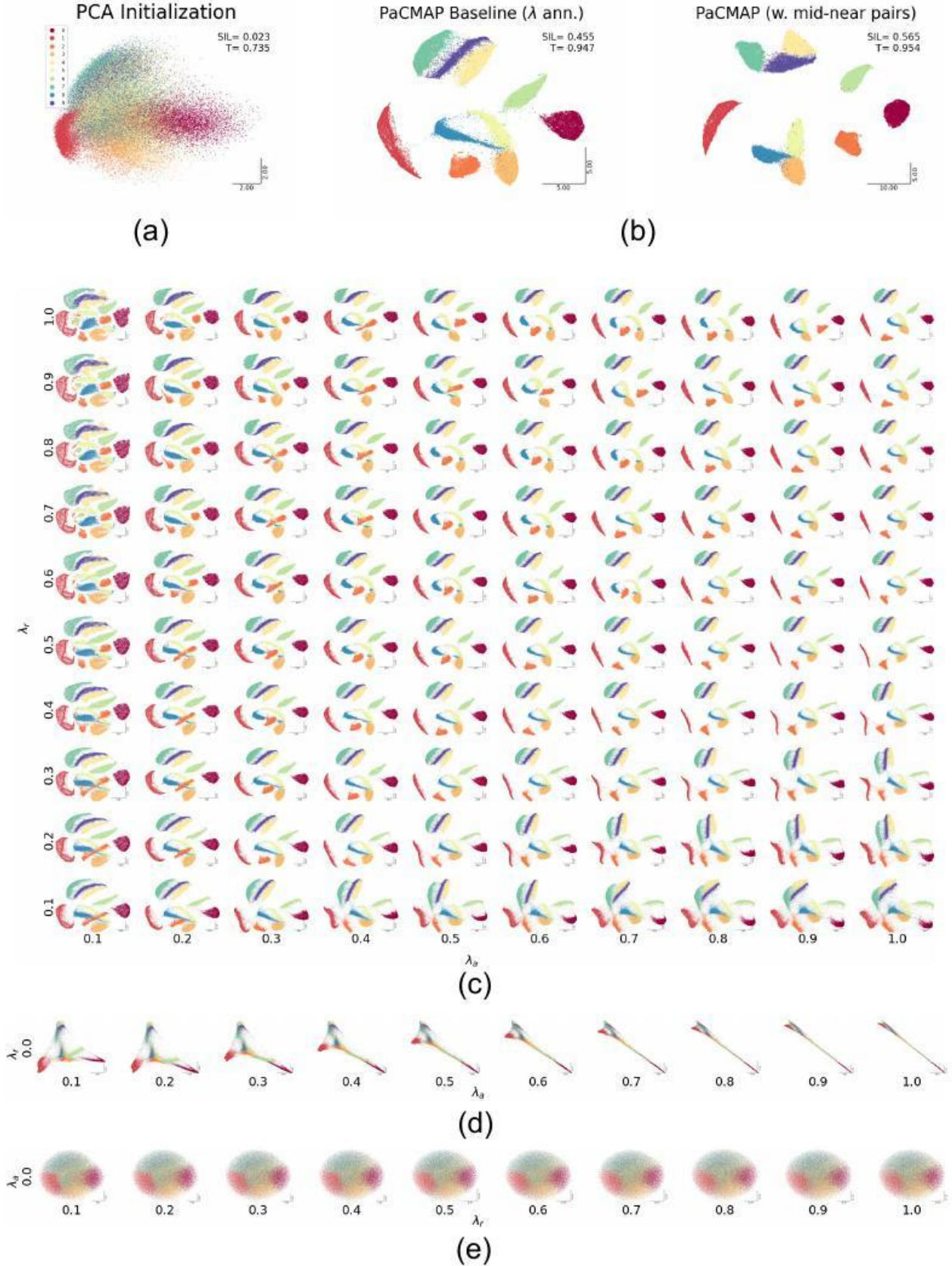


Figure 22: Varying λ_a and λ_r for the MNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Left: baseline with λ annealed from 1, with mid-near points excluded so as to isolate the attraction–repulsion interaction; right: with mid-near points included. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

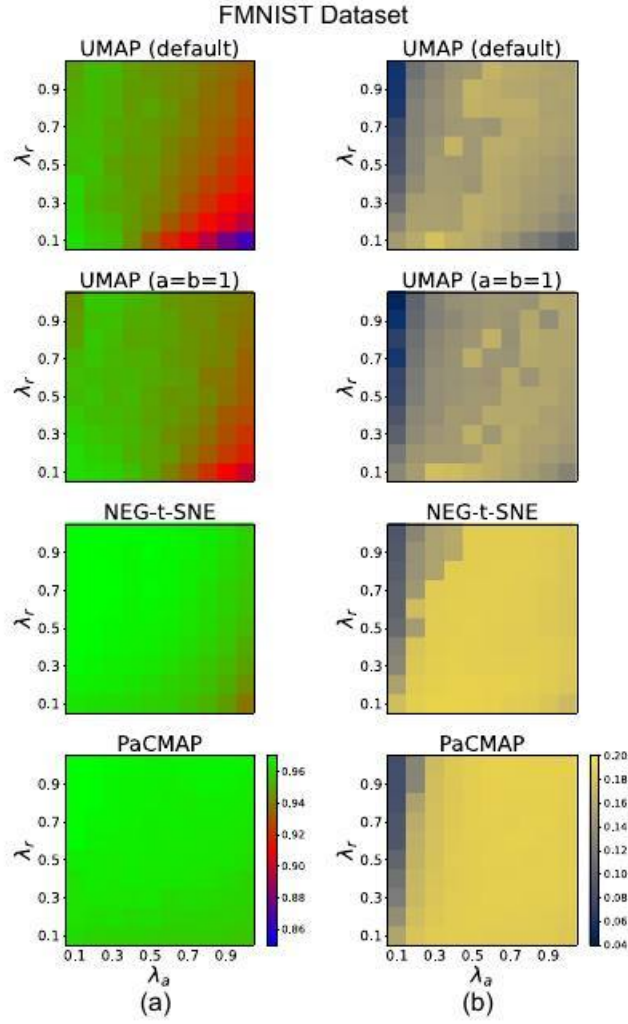


Figure 23: (a) Trustworthiness and (b) Silhouette score across methods for the FMNIST dataset, as λ_a and λ_r are varied independently.

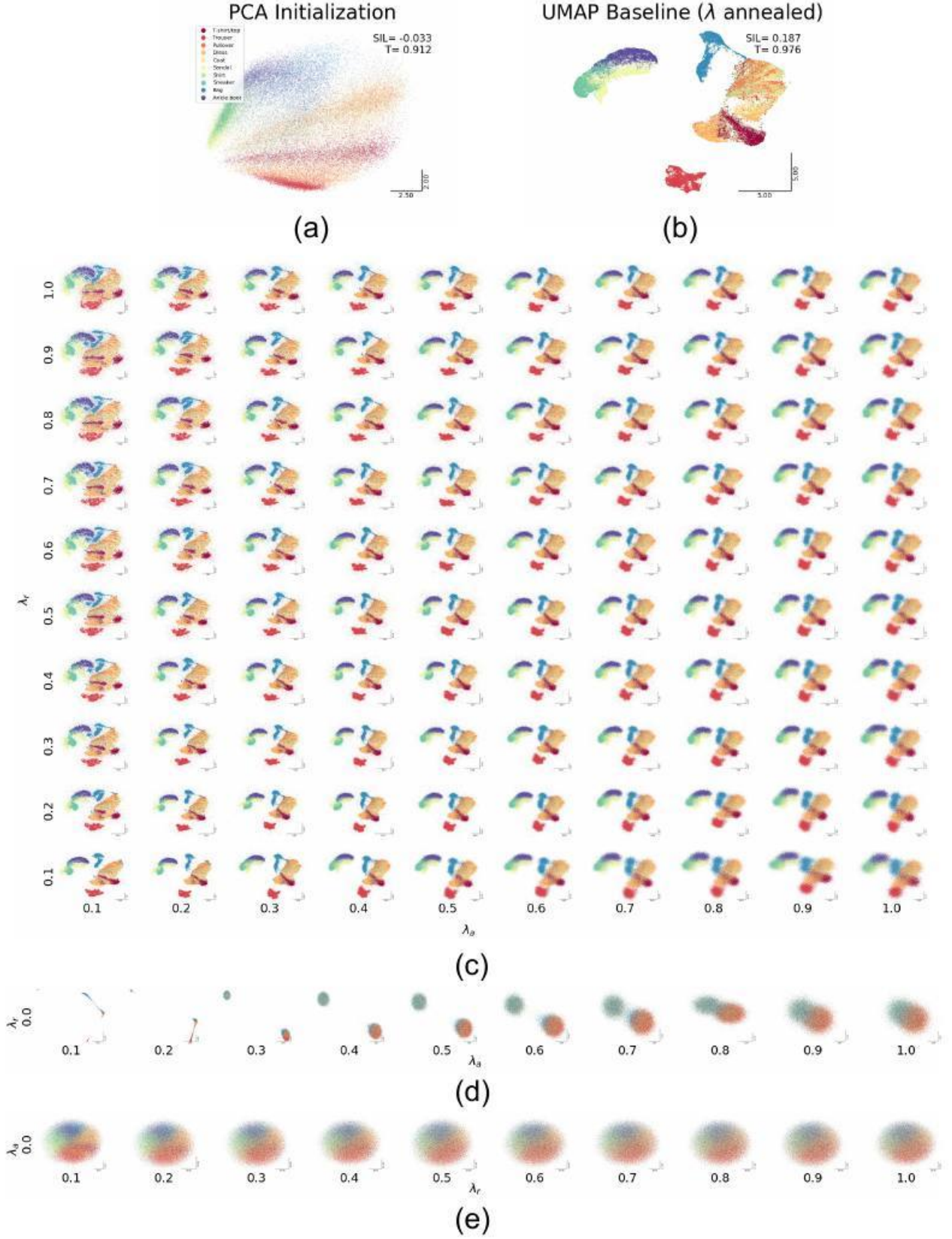


Figure 24: Varying λ_a and λ_r for the FMNIST dataset using UMAP's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline with λ annealed from 1. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

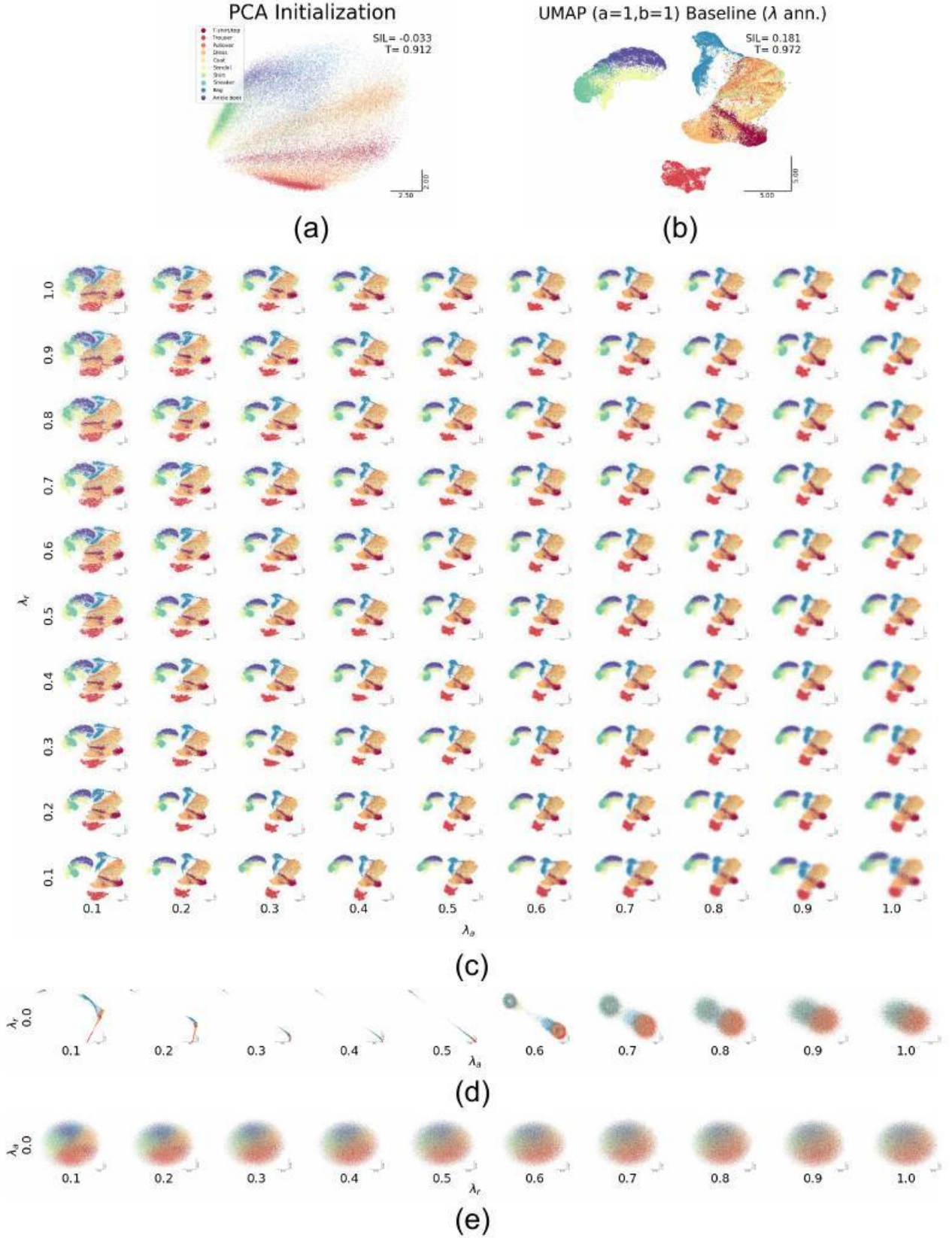


Figure 25: Varying λ_a and λ_r for the FMNIST dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization of the embeddings. (b) Baseline with λ annealed from 1. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

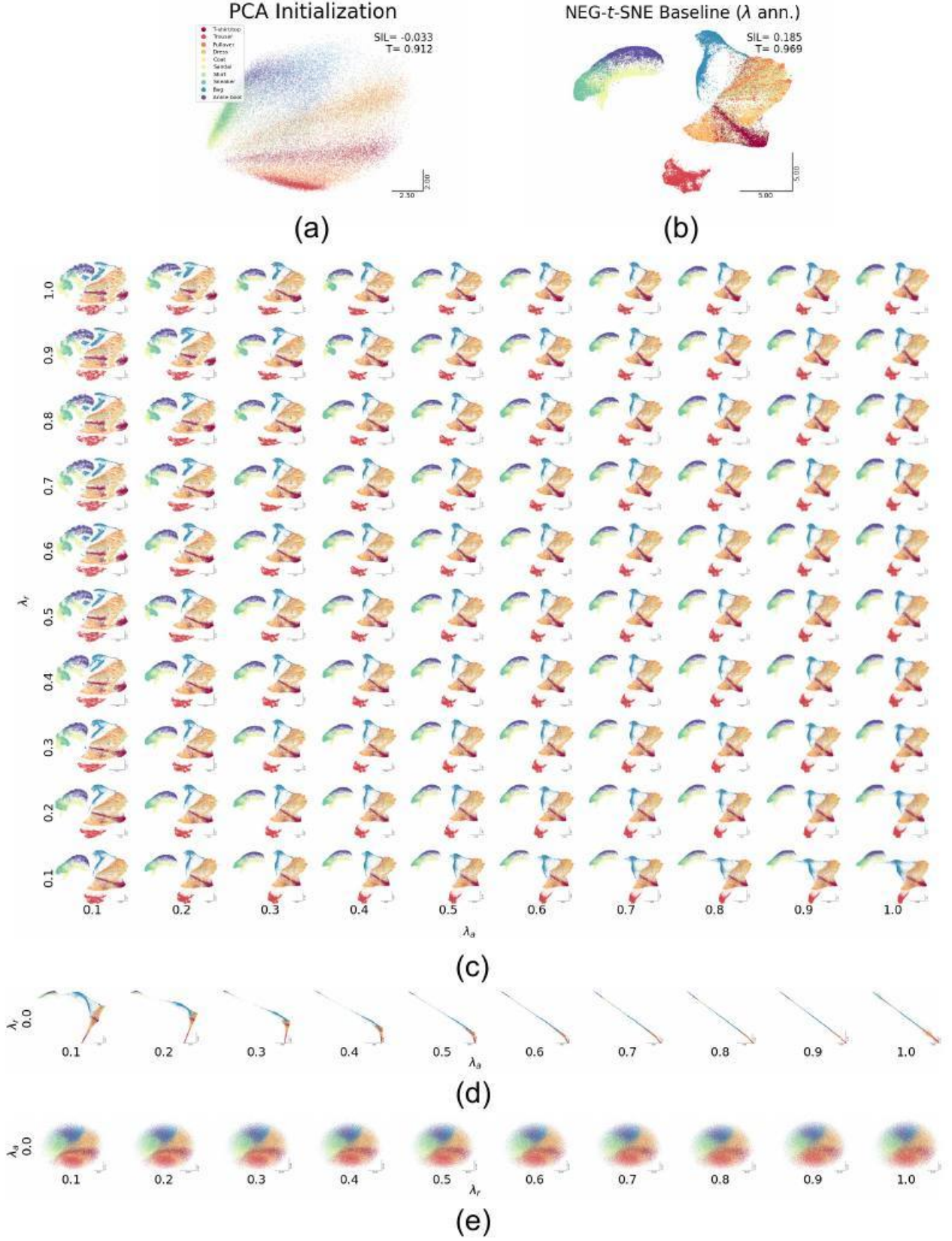


Figure 26: Varying λ_a and λ_r for the FMNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline with λ annealed from 1. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

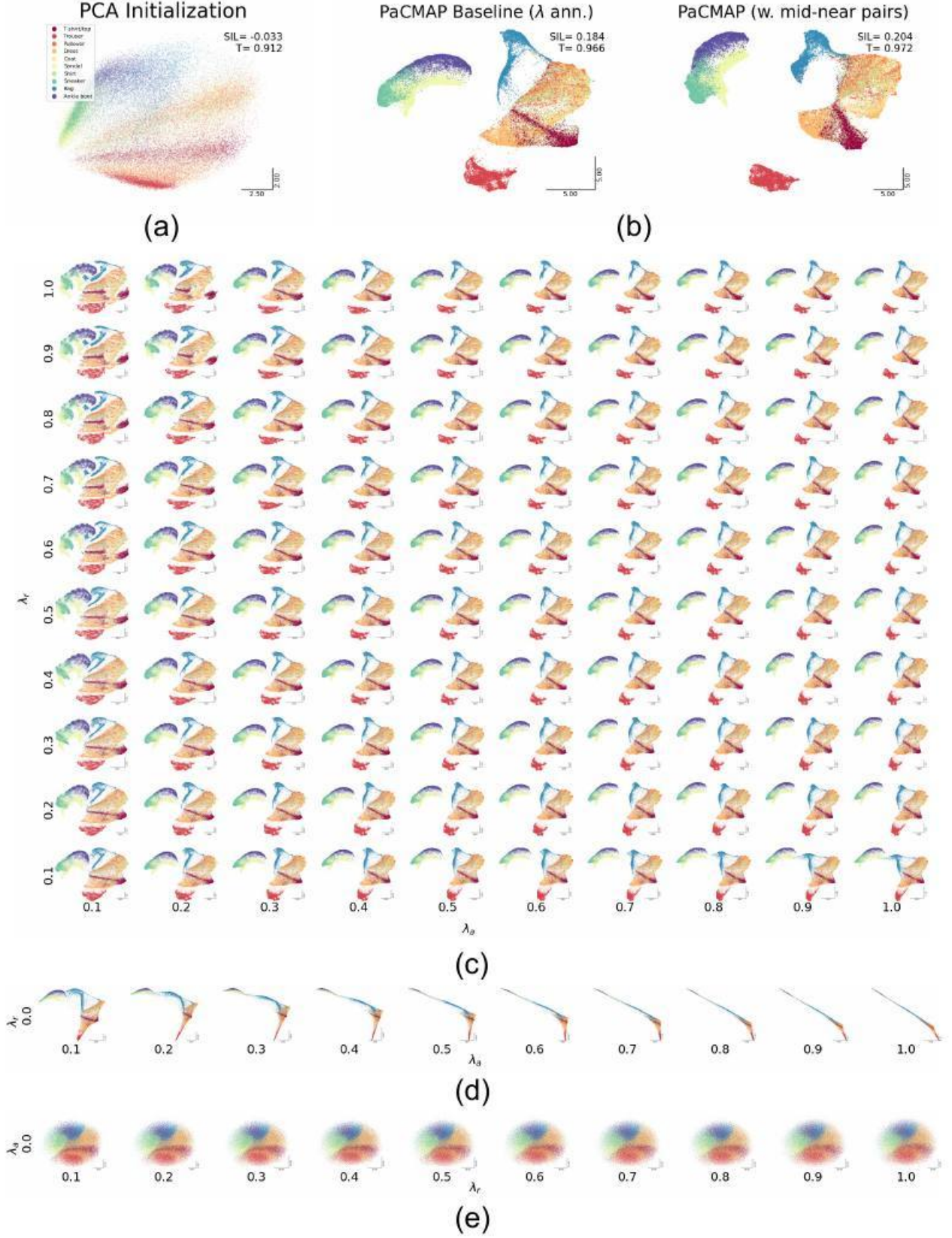


Figure 27: Varying λ_a and λ_r for the FMNIST dataset using PaCMAP’s attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Left: baseline with λ annealed from 1, with mid-near points excluded so as to isolate the attraction–repulsion interaction; right: with mid-near points included. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

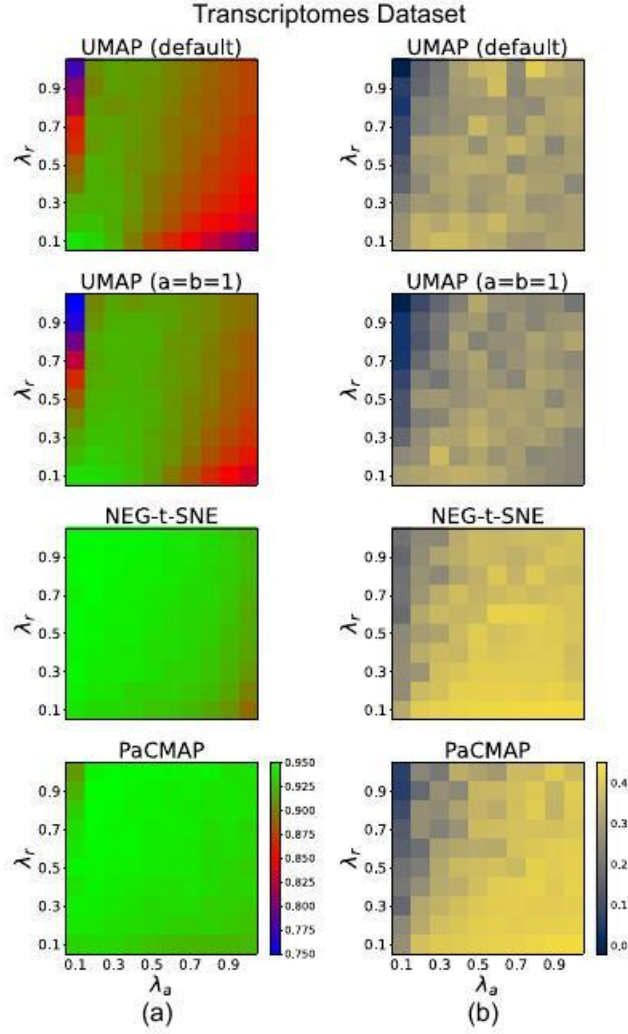


Figure 28: (a) Trustworthiness and (b) Silhouette score across methods for the single-cell transcriptomes dataset, as λ_a and λ_r are varied independently.

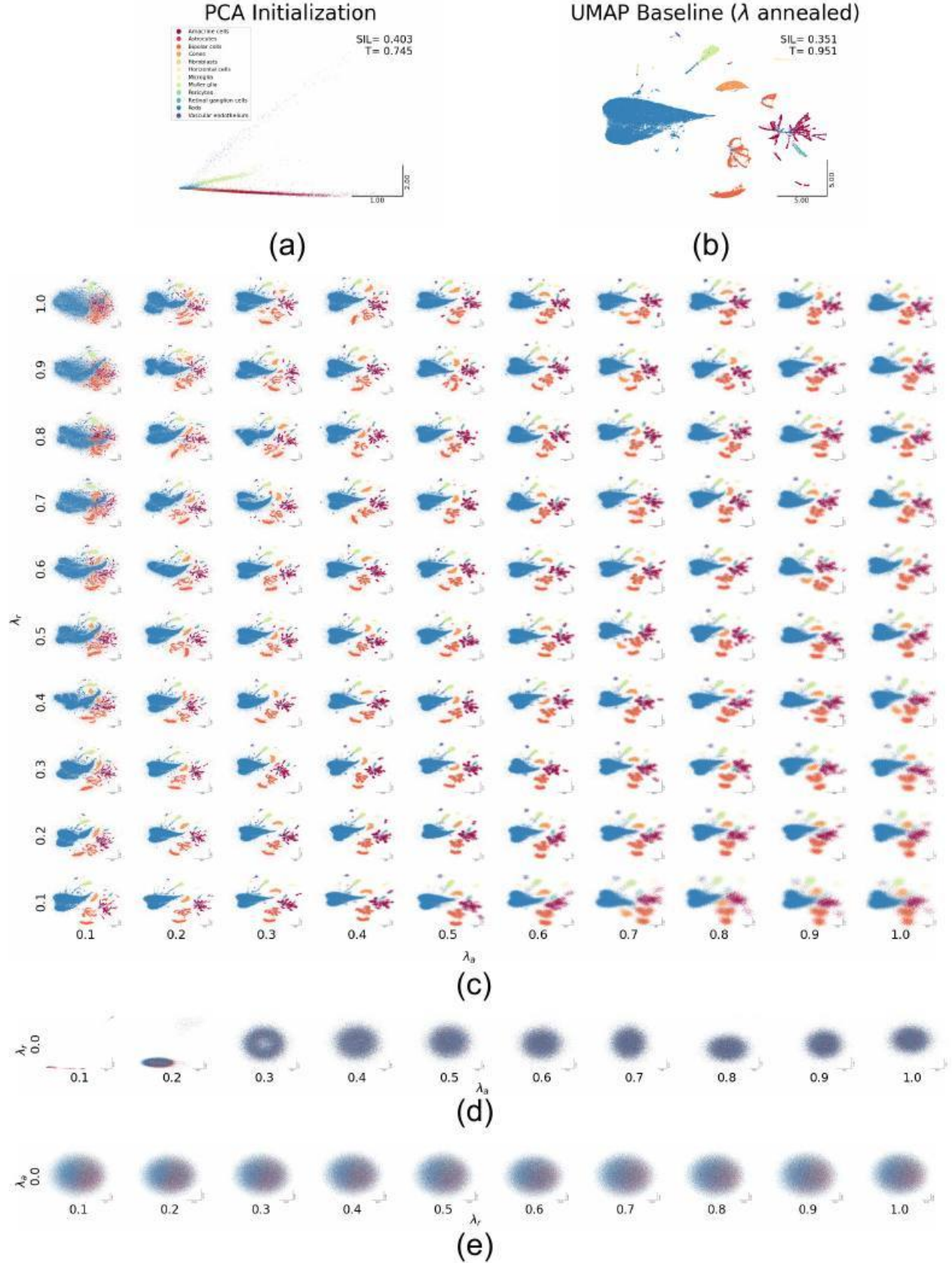


Figure 29: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline with λ annealed from 1. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

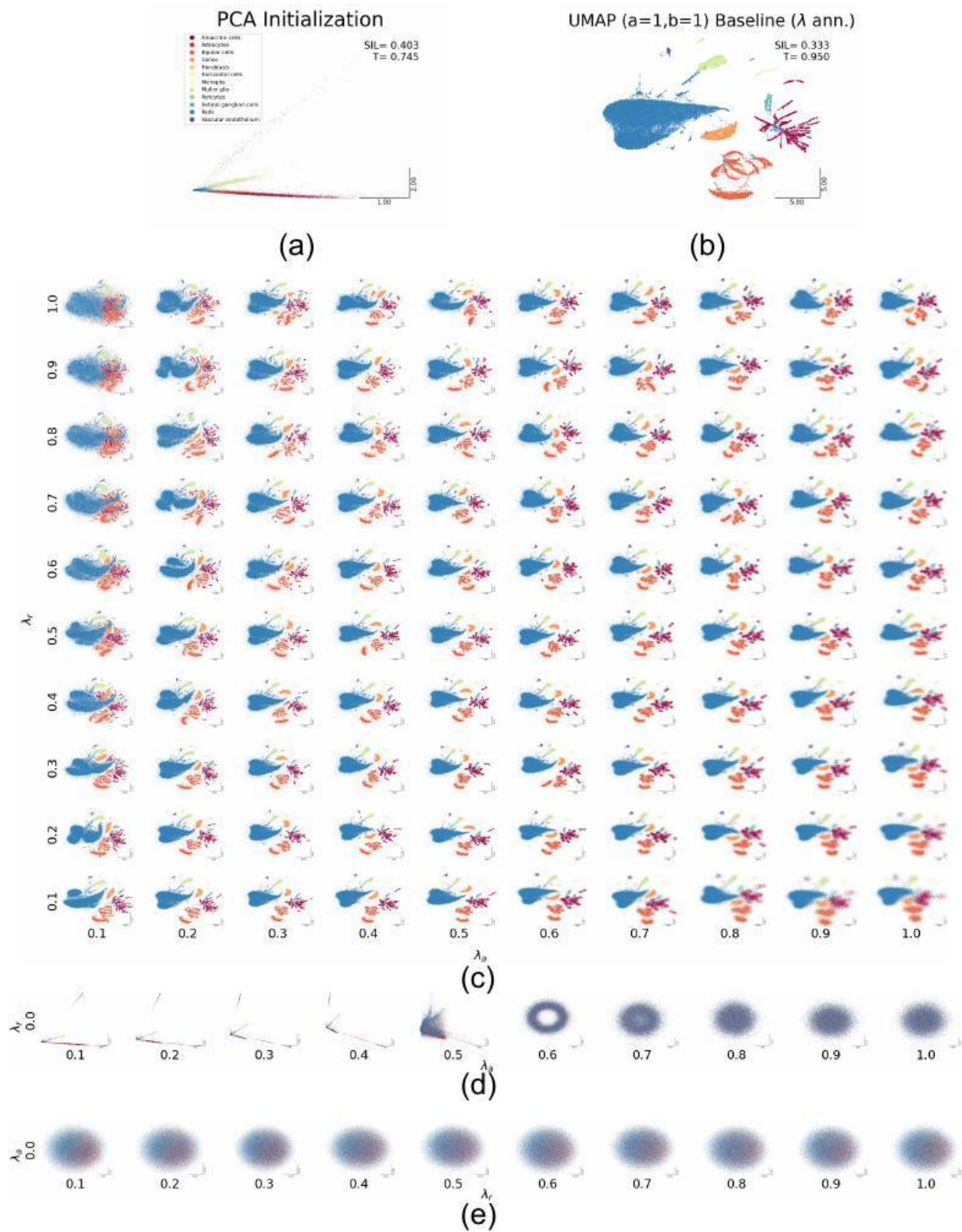


Figure 30: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), repulsive force alone cannot produce any clusters.

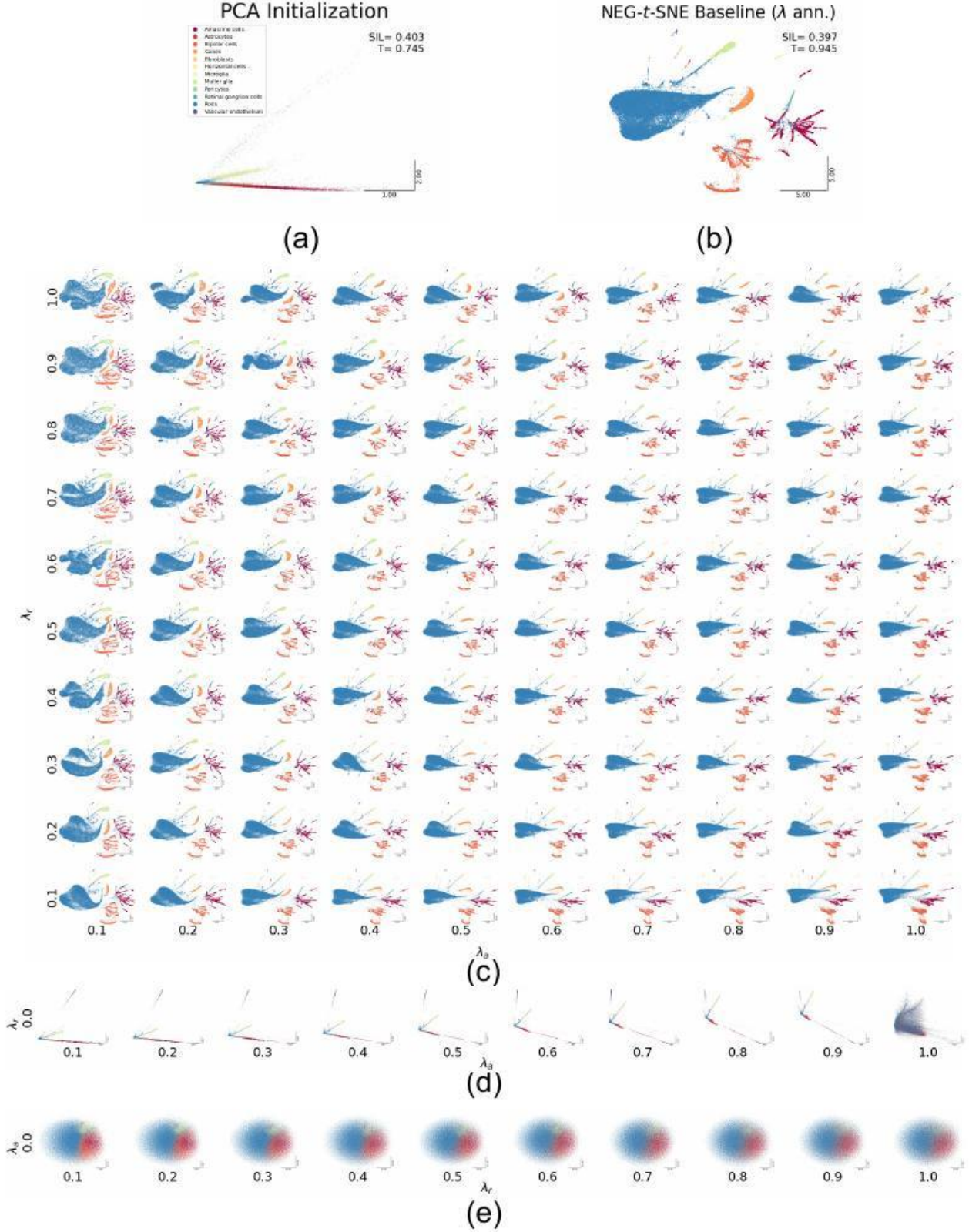


Figure 31: Varying λ_a and λ_r for the single-cell transcriptomes dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline with λ annealed from 1. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no distinct clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

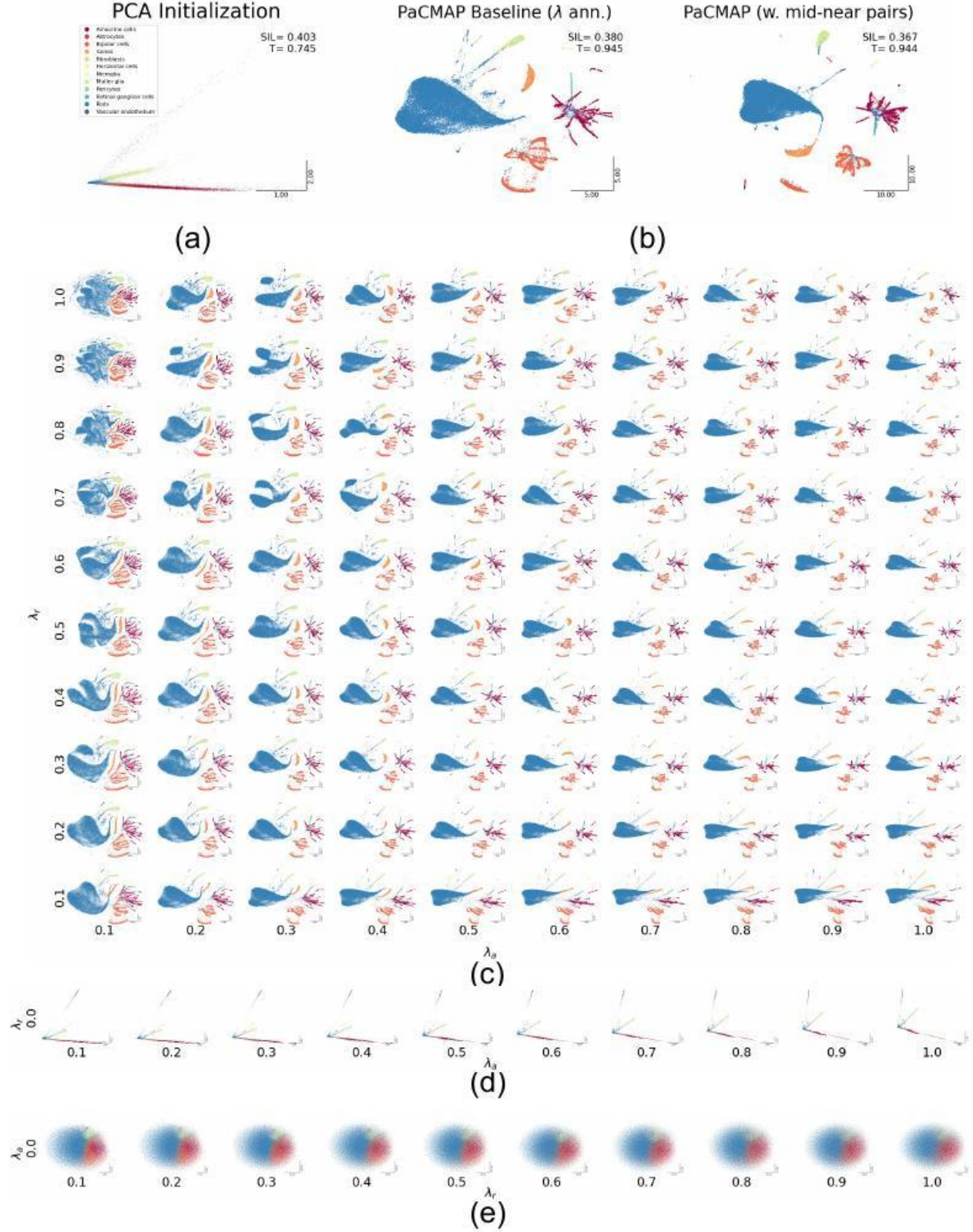


Figure 32: Varying λ_a and λ_r for the single-cell transcriptomes dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Left: baseline with λ annealed from 1, with mid-near points excluded so as to isolate the attraction–repulsion interaction; right: with mid-near points included. (c) Embeddings as λ_a and λ_r vary, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Together, (d) and (e) confirm that both forces are indispensable to cluster formation.

J Mix and Match Approach

Because our analysis treats the attraction and repulsion shapes as mutually independent, it directly implies that shapes derived from different loss functions and principles can be freely recombined. Figure 33 presents several such combinations. To this end, we confined the attraction shapes to $[-1, 0]$ (so that Proposition 4.1 holds and the shapes remain insensitive to the learning rate) and paired them with finite repulsion shapes (thereby suppressing the large repulsive forces that arise at small distances). We held the learning rate constant at $\lambda = 1$ and used UMAP’s optimization throughout, so that any variation is attributable to the shapes alone rather than to learning-rate scheduling or the optimizer. The results delineate a clear trade-off: UMAP’s attraction shape (via $f_a = 0.5f_a^U$) yields better clustering (higher silhouette score), whereas PaCMAP’s attraction shape yields better local structure (higher trustworthiness), the latter because PaCMAP’s shape lies within $[-0.5, 0]$ and thus admits no position flips during the updates. Confining the attraction shapes of UMAP and NEG- t -SNE to $[-0.5, 0]$ (via $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively) improves local structure at the expense of clustering performance (Fig. 34).

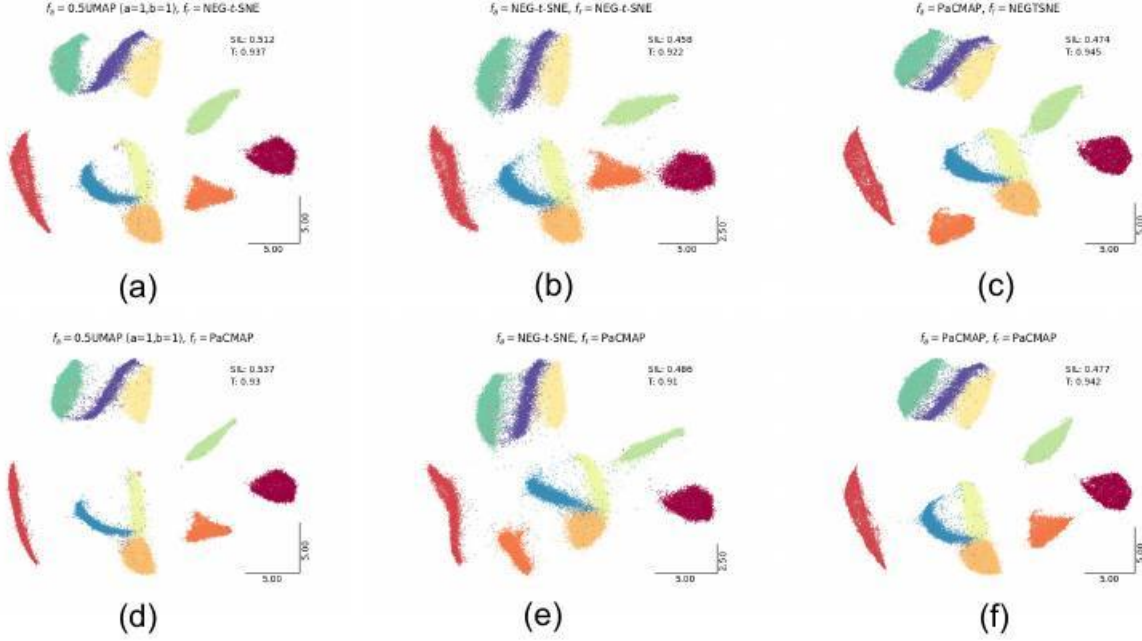


Figure 33: Embeddings of MNIST obtained by mixing and matching attraction and repulsion shapes at a constant learning rate ($\lambda = 1$), using attraction shapes that satisfy Proposition 4.1 and finite repulsion shapes. Top row: the attraction shapes of UMAP, NEG- t -SNE, and PaCMAP, each paired with NEG- t -SNE’s repulsion shape. Bottom row: the same attraction shapes, each paired with PaCMAP’s repulsion shape.

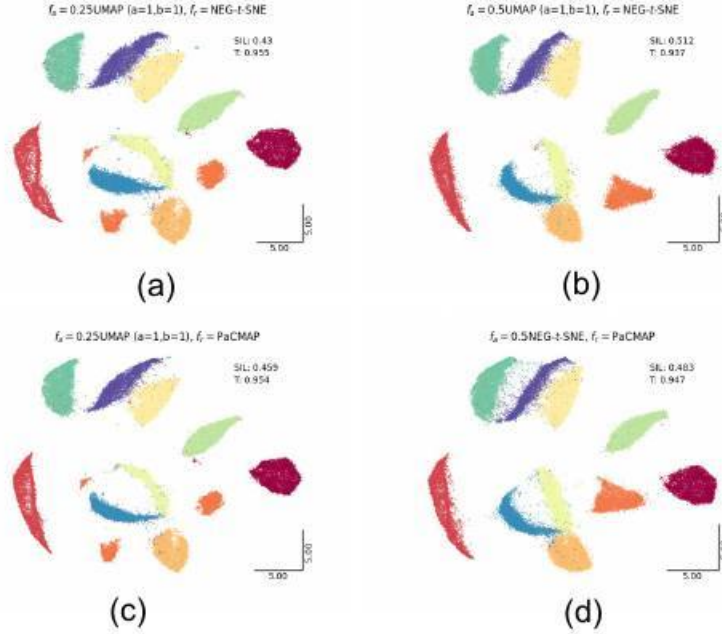


Figure 34: Embeddings of MNIST obtained by mixing and matching attraction and repulsion shapes at a constant learning rate ($\lambda = 1$), with the attraction shapes confined to $[-0.5, 0]$ to preclude any flips during the attractive update. Top row: the attraction shapes of UMAP and NEG- t -SNE, each paired with NEG- t -SNE’s repulsion shape. Bottom row: the same attraction shapes, each paired with PaCMAP’s repulsion shape.

K Implementation Details

For analysis, we implemented our own UMAP algorithm. We used `numba` (Lam et al., 2015) to compute an exact nearest neighbor graph (instead of an approximate one) with $k = 15$ and `scikit-learn`'s (Pedregosa et al., 2011) implementation of the PCA algorithm for PCA initialization. We used this to produce and quantify the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9 and 10. We also used the same implementation when we changed the attraction and the repulsion shapes to those of the alternative methods (Figs. 18, 34, unless otherwise stated). The trustworthiness and silhouette scores were computed using the corresponding function from the `scikit-learn` package.

The mappings shown in Figs. 2, 3, 9, 10 and 15 are rotated to a reference embedding ((a) for each respective Figures). To achieve this, we performed Procrustes alignment of the embeddings by normalizing them (zero mean and unit norm) and then using SciPy's (Virtanen et al., 2020) `orthogonal_procrustes` method to extract rotation and scaling parameters.

To compare with Neg- t -SNE in Fig. 11, we used the original implementation of the contrastive embedding framework for both the UMAP and the Neg- t -SNE algorithms; the implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22(b) (right), 27(b) (right), and 32(b) (right) were obtained using the official PaCMAP package; the implementation details will be released upon acceptance.

The codes for displaying the embeddings of the PubMed dataset in Fig. 7 will be released upon acceptance.